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Detecting Non-Equilibrium Quasiparticle Bursts in Granular Aluminium Resonators

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The work presented in this Thesis is, to the best of my knowledge and belief original, except as acknowledged in the text, and has not been submitted either in whole or in part, for a degree at this or any other university.

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Abstract

Non-equilibrium quasiparticle bursts can occur in superconducting devices when high energy ionising radiation impacts the chip on which superconducting devices sit on, producing high energy phonons which propagate through the device. Quasiparticles, which can be thought of as broken Cooper pairs, dissipate energy from superconducting devices and are a source of decoherence in superconducting qubits. Time-correlated quasiparticle bursts also occur in multiple devices on the same chip and last for several seconds after the initial event. This is a significant problem for high kinetic inductance detectors and quantum error correction protocols. We use high kinetic inductance granular aluminium resonators fabricated onto a silicon chip to detect time-correlated quasiparticle bursts in two resonators on the same chip. Quasiparticle bursts are found to be randomly distributed with a Poisson distribution, and we measure the average quasiparticle burst rate to be one every 618 s. The quasiparticle decay times are measured to be on the order of half a second in our resonators. This thesis is part of a larger project to reduce the quasiparticle burst rate by fabricating a structure that will act as a phononic low pass filter.

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I would like to thank Dr. Eugene Sachkou, for the opportunity to work on his project involving superconducting qubits. I got to experience a whole other side of this field of research, and gained valuable knowledge and experience by performing measurements on his device. I also thank him for taking the time to help me understand the measurements and the measurement process, a skill that I can take with me going forward.

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1
Introduction

Quantum computers are an emerging technology that aims to exploit properties of the many-particle wavefunction, such as the superposition of states and the entanglement of quantum systems, to solve a set of computational problems that are highly inefficient on classical computers [8]. As such, quantum computers have potential applications in several important fields of research, such as computational chemistry [9] and cybersecurity [10], as well as applications in industry, such as solving transportation problems in logistics systems [11]. The fundamental component of any quantum processor is the quantum bit, or qubit. There are many promising qubit designs, with some of the most notable using linear optics and neutral atoms [12][13]. One of the most promising qubit designs however comes in the form of superconducting circuits [14]. Superconducting qubits have three main advantages over other qubit designs. Firstly, unlike many other qubit designs, superconducting qubits can have their properties, such as the spacing of the energy levels used to store information, decided upon during fabrication. This means that they can be fabricated to be more or less resilient to certain types of noise, allowing researchers to design their qubits to best match their needs [15]. Secondly, since superconducting qubits are macroscopic systems, they can be fabricated using current manufacturing technology already present in the semiconductor industry. Lastly, superconducting qubits can be very precisely controlled and easily entangled [16]. However, due to their macroscopic size, superconducting qubits couple very strongly to their environment and are therefore highly sensitive to noise.

Superconducting qubits have traditionally exhibited very low coherence times compared to other designs. The first implementation of a superconducting qubit, the Cooper pair box, had a coherence time on the order of a few nanoseconds [17]. Research over the years has been focused on improving the coherence times of superconducting qubits through new designs and better ways to isolate the system from the environment [18]. One of the most promising designs of modern qubits is the transmon qubit, which consists of a Josephson junction connected in parallel with a shunt capacitor. This creates an anharmonic oscillator circuit whose energy levels can be used as qubit states. Modern transmon qubits have been shown to experience coherence times over 100 μs [19].

One of the factors limiting superconducting circuits is generation of non-equilibrium quasiparticles in the superconducting elements of the qubit. A material becomes superconducting when the temperature becomes low enough for electrons to pair together into Cooper pairs which can flow without resistance. Quasiparticles can be viewed as broken Cooper pairs and degrade the performance of superconducting circuits in two ways: their presence introduces dissipation and fluctuations in their numbers give rise to noise [4]. The density of quasiparticles observed in these superconducting circuits is a lot higher than what is theoretically predicted at thermal equilibrium by the Bardeen-Cooper-Schrieffer theory of superconductivity [5]. It has recently been shown that these non-equilibrium quasiparticles are generated when high energy ionising radiation impacts the chip that the qubit sits on. These high energy impacts cause high energy phonons to propagate throughout the chip and break Cooper pairs in superconducting devices on the chip. These quasiparticles contribute to qubit decoherence when they tunnel across Josephson junctions, a process known as quasiparticle poisoning [4] [5].

This result is shown in two main papers published in May of 2020 and August of 2020 respectively. The first paper, (Cardani et al, 2020), used high inductance granular aluminium resonators to accurately detect quasiparticle events in three resonators on the same chip and showed that the quasiparticle events are time correlated across resonators, which implies that the energy is generated in the silicon substrate that the resonators sit on and is transported to the resonators by means of phonons [4]. They also showed that by increasing the amount of radiation shielding around the resonators, they could reduce the number of time correlated quasiparticle events detected, which implies that the phonons are being generated by radiation interacting with the chip. The second paper (Vepsalainen et al, 2020) used a chip containing two transmon qubits and showed the effect of ionising radiation on qubit coherence time [5].

This thesis is part of a larger project in the Superconducting Quantum Devices Laboratory, in collaboration with the Bowen group, at the University of Queensland. The project aims to suppress non-equilibrium quasiparticle bursts in granular aluminium resonators by fabricating them on a trampoline mechanical resonator structure, which will act as a phononic low pass filter [20]. The aim of this thesis is to design the experimental apparatus in which the chip is placed in, to use high inductance granular aluminium resonators to detect non-equilibrium quasiparticle bursts in at least two resonators on the same chip and to determine the quasiparticle burst rate in our devices before the trampoline mechanical resonator structure is introduces.

1.1 Structure of this thesis

The structure of this thesis is as follows. Chapter 2 will provide an introductory background to the electromagnetic theory required for the simulation and design of the waveguide used. Chapter 3 will motivate the theory behind superconductivity and what quasiparticles are in terms of Bardeen-Cooper-Shrieffer theory, as well as introduce the tools used to characterise our resonators before we can detect quasiparticle bursts. Chapter 4 is a literature review of the work done since May of 2020 that has improved our understanding of non equilibrium quasiparticles and illustrated why non-equilibrium quasiparticles are an important problem to overcome. Chapter 5 contains information on the finite element simulations done to design the waveguide, as well as the design choices. Chapter 6 describes the experimental methodology, and a brief discussion on how quasiparticles are detected. Chapter 7 goes through the analysis of the data and summarises the results.

2
Electromagnetism

Our experiment consists of granular aluminium resonators fabricated onto a silicon chip which is eventually placed in a waveguide. In this chapter, I will give a theoretical overview of the electromagnetic theory used in the design of the experiment and the analysis of the data.

2.1 Waveguides

The chip is designed to fit inside an aluminium waveguide with two SMA couplers which act as input and output ports. A waveguide is a hollow conductor that allows electromagnetic waves to propagate through it. For a general waveguide, we can assume that the waveguide is a perfect conductor, so that the electric field E and the magnetic field B are set to zero inside the material itself, and the boundary conditions of the inner wall are

$$E_{\parallel} = 0 \tag{2.1}$$
$$B_{\perp} = 0. \tag{2.2}$$

Monochromatic waves that propagate down the tube will have the generic form [1]

$$E(x, y, z, t) = E_0(x, y, z)e^{i(kz-\omega t)} \tag{2.3}$$
$$B(x, y, z, t) = B_0(x, y, z)e^{i(kz-\omega t)}, \tag{2.4}$$

where z is the direction of propagation, k is the wavenumber and ω is the angular frequency of the wave. The electric and magnetic fields in the interior of the waveguide will also satisfy

Maxwell's equations,

$$\nabla \cdot E = 0 \tag{2.5}$$
$$\nabla \cdot B = 0 \tag{2.6}$$
$$\nabla \times E = -\frac{\partial B}{\partial t} \tag{2.7}$$
$$\nabla \times B = \frac{1}{c^2}\frac{\partial E}{\partial t}, \tag{2.8}$$

with initial conditions

$$E_0(x, y, z) = E_x\hat{x} + E_y\hat{y} + E_z\hat{z} \tag{2.9}$$
$$B_0(x, y, z) = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}. \tag{2.10}$$

The fields can be expressed as column vectors, with relationships between each component being determined by Eqs. (2.7, 2.8) to be

$$\begin{pmatrix} \partial_y E_z - ikE_y \\ ikE_x - \partial_x E_z \\ \partial_x E_y - \partial_y E_x \end{pmatrix} = i\omega \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \tag{2.11}$$
$$\begin{pmatrix} \partial_y B_z - ikB_y \\ ikB_x - \partial_x B_z \\ \partial_x B_y - \partial_y B_x \end{pmatrix} = -\frac{i\omega}{c^2} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}. \tag{2.12}$$

The x and y components of the electric and magnetic field vectors can then be written in terms of the z components as,

$$E_x = \frac{i}{(\omega/c)^2 - k^2}\left(k\partial_x E_z + \omega\partial_y B_z\right) \tag{2.13}$$
$$E_y = \frac{i}{(\omega/c)^2 - k^2}\left(k\partial_y E_z - \omega\partial_x B_z\right) \tag{2.14}$$
$$B_x = \frac{i}{(\omega/c)^2 - k^2}\left(k\partial_x B_z - \frac{\omega}{c^2}\partial_y B_z\right) \tag{2.15}$$
$$B_y = \frac{i}{(\omega/c)^2 - k^2}\left(k\partial_y B_z + \frac{\omega}{c^2}\partial_x E_z\right). \tag{2.16}$$

Lastly, using Eqs. (2.5, 2.6), the following relations for the z components of the electric and magnetic fields are found

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] E_z = 0 \tag{2.17}$$
$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] B_z = 0. \tag{2.18}$$

If $E_z = 0$, the waves are said to be transverse electric (TE), and if $B_z = 0$, the waves are said to be transverse magnetic (TM) waves. If both $E_z = 0$ and $B_z = 0$, the waves are said to be TEM waves. Since TEM waves cannot exist within a waveguide, they are not considered further [1].

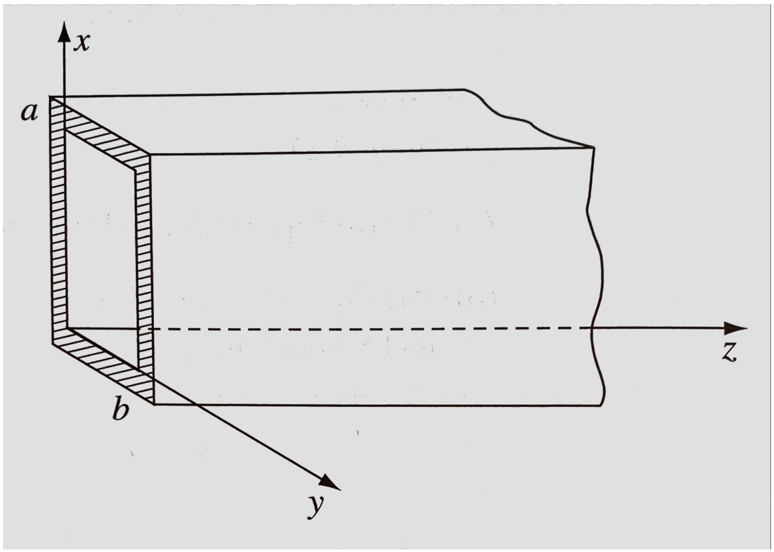

FIGURE 2.1: Diagram of a rectangular waveguide of height a and width b [1]

2.1.1 TE modes in a rectangular waveguide

Consider a rectangular waveguide of height a and width b. Since we are dealing with TE modes, $E_z = 0$. The solution to B_z can be found using a separation of variables approach. Let

$$B_z(x, y) = X(x)Y(y). \tag{2.19}$$

Substituting into the TE mode condition in Eq. (2.18),

$$Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + ((\omega/c)^2 - k^2)XY = 0 \tag{2.20}$$

$$\implies \frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + (\omega/c)^2 - k^2 = 0. \tag{2.21}$$

Since each term in the result is independent, they are each constants. Therefore, we can choose constants k_x and k_y such that

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -k_x^2 \tag{2.22}$$

$$\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -k_y^2. \tag{2.23}$$

The general solution for X is therefore

$$X(x) = A\sin(k_x x) + B\cos(k_x x). \tag{2.24}$$

Applying the boundary conditions, which require that B_x and its derivative to vanish at $x = 0$ and $x = a$, we find that $A = 0$ and $k_x = m\pi/a$. A similar argument is made for Y, where $k_y = n\pi/b$. Therefore, the magnetic field

$$B_z = B_0 \cos(m\pi x/a)\cos(n\pi y/b) \tag{2.25}$$

characterises the TE_{mn} mode. Substituting Eqs. (2.23, 2.22) into Eq. (2.21), the total wavenumber which characterises the wave can be found to be

$$k^2 = (\omega/c)^2 - k_x^2 - k_y^2 \tag{2.26}$$

$$\implies k = \sqrt{(\omega/c)^2 - (m\pi/a)^2 - (n\pi/b)^2}. \tag{2.27}$$

If the frequency of the wave is small enough (that is, when $(\omega/c)^2 < (\pi m/a)^2 + (\pi n/b)^2$), the wavenumber is imaginary, meaning that the wave will decay exponentially as it passes through the waveguide. Hence, we define the lowest frequency at which the wave will propagate as the cutoff frequency [1]

$$\omega_{mn} = c\pi\sqrt{(m/a)^2 + (n/b)^2}. \tag{2.28}$$

2.2 Scattering parameters

To obtain the transmitted or reflected power in a circuit network, such as a resonator coupled to a waveguide, the voltage coming out of a port is compared to the voltage going into the port. For a general N-port network, the scattering matrix, or S-matrix, is defined as

$$\begin{pmatrix} V_1^- \\ V_2^- \\ \vdots \\ V_N^- \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} & \cdots & S_{1N} \\ S_{21} & S_{22} & \cdots & S_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ S_{N1} & S_{N2} & \cdots & S_{NN} \end{pmatrix} \cdot \begin{pmatrix} V_1^+ \\ V_2^+ \\ \vdots \\ V_N^+ \end{pmatrix} \tag{2.29}$$

where V_i^- is the output voltage from port i and V_j^+ is the input voltage for port j [21]. For a single entry of the S-matrix, the scattering parameter is

$$S_{ij} = \frac{V_i^-}{V_j^+} \tag{2.30}$$

when all other input ports are set to zero, ie $V_k^+ = 0\ \forall k \neq j$ .For a two port network, like a waveguide, S_{11} measures the amount of power generated at port 1 reflected back by the system, and S_{21} measures the amount of power generated at port 1 that is transmitted to port 2. The transmission spectrum holds the information about the system, such as the resonator frequency and line width. The scattering parameters are usually represented in terms of the gain through the system, which is defined as [21]

$$Gain = 20log_{10}(S_{ij})dB. \tag{2.31}$$

where dB are the units for decibels. For a two port network, the scattering parameters S_{21} and S_{11} are sufficient fully describe the network.

3

Superconducting Resonators

This chapter continues the discussion in the relevant background knowledge required for the experiment. Here, I motivate the theory of superconductivity and quasiparticles in terms of Bardeen-Cooper-Shrieffer (BCS) theory. I then discuss a derivation of the transmission parameter for a resonator in a waveguide and discuss a fitting routing used to determine our resonators parameters.

3.1 Brief theory on superconductors

3.1.1 Superconductivity

Superconductivity is a macroscopic phenomenon that occurs when a material is cooled below the materials superconducting critical temperature T_C, provided that it is not subject to magnetic fields greater than the materials critical magnetic field strength B_C. When electrons flow through an ionic lattice, the lattice responds to that motion in a way that results in a net attractive interaction term between electrons in the lattice whose energies are sufficiently close together. In isolation, this attractive interaction between two electrons would be negligible, but in 1956, Cooper showed that in the presence of additional electrons in the Fermi sphere, the exclusion principle alters the two electron problem such that a bound state exists even for weak interactions [22][23]. The bound states are called Cooper pairs and will form when the temperature of the material is low enough. When this condition is met, the Fermi sea is unstable, meaning that electrons in the material will pair up until an equilibrium point is reached [24]. BCS theory is an extension on the theory derived by Cooper which constructs a ground state wavefunction in which all electrons in the material have paired together to form Cooper pairs[22]. Cooper pairs have the unique property that they can be conducted through the material with zero electrical resistance. Furthermore, when a superconductor is placed in an external magnetic field, a supercurrent

J_s arises in the material, which shields magnetic fields below a certain strength, leading to perfect diamagnetism. This is known as the Meissner effect [25]. The supercurrent J_s is given by

$$J_s(r,t) = 2e\Re\left[\psi^*\left(\frac{\hbar}{m_s i}\nabla - \frac{2e}{m_s}\mathbf{A}(r,t)\right)\right] \tag{3.1}$$

$$= \frac{2en_s\hbar}{m_s}\left(\nabla\chi(\mathbf{r},t) - \frac{2\pi}{\Phi_0}\mathbf{A}(\mathbf{r},t)\right), \tag{3.2}$$

where $\Re$ denoted the real part of the expression, $2e$ is the charge of a Cooper pair with mass m_s, $\phi_0 = h/2e$ is the flux quantum, and A is the vector potential, which is defined such that $B(r,t) = \nabla \times A(r,t)$. If the magnetic field exceeds B_C, the superconducting state will collapse. Type I and type II superconductors can be distinguished based on how the superconducting state collapses in the presence of large magnetic fields. Type I superconductors will simply collapse, while type II superconductors allow the field to penetrate the material in the form of flux holes until the state collapses at a magnetic field strength of $B_{C,2}$. Granular Aluminium is a type II superconductor with critical field strength of up to 3.6 T [26].

3.1.2 Quasiparticles

BCS theory can be described in second quantisation by the diagonalised Hamiltonian

$$\mathcal{H} = \sum_k (\epsilon_k - E_k + \Delta_k b_k^*) + \sum_k E_k (\gamma_{k0}^*\gamma_{k0} + \gamma_{k1}^*\gamma_{k1}), \tag{3.3}$$

where ϵ_k is the energy of an electron with momentum k in a Fermi sea, Δ_k is the minimum excitation energy, $E_k = (\Delta_k^2 + \xi_k^2)^{\frac{1}{2}}$ is the excitation energy of a quasiparticle with momentum $\hbar k$, $\xi_k = \epsilon_k - E_F$ (where E_F is the Fermi energy) and $b_k = \langle c_{-k\downarrow} c_{k\uparrow}\rangle$ where $c_k\sigma$ is the annihilation operator for an electron. γ_k^* and γ_k are creation and annihilation operators for quasiparticles. The first term of the Hamiltonian is constant and corresponds to the energy of the BCS ground state. The second term gives the increase in the energy above the ground state in terms of quasiparticle number operators $\gamma_k^*\gamma_k$. Therefore, in BCS theory, quasiparticles are excitations on the BCS ground state with an excitation energy E_k [24].

Phenomenologically, quasiparticles are essentially unpaired electrons in a superconductor, and are governed by Fermi-Dirac statistics [27]. Accordingly, the expected number of quasiparticles per unit volume in a superconductor where $k_BT < \Delta_k$ is given by

$$n_{qp} = 2N_0\sqrt{2\pi k_B T \Delta_k}\exp(-\Delta_k/k_BT), \tag{3.4}$$

where N_0 is the single spin density of states at the Fermi level, k_B is the Boltzmann constant and T is the temperature [28]. As the temperature approaches zero, we would expect that the number of quasiparticles decays exponentially towards zero as well. However, the number of quasiparticles tends to plateau at a non-zero value instead, deviating from the thermal equilibrium prediction [29]. These extra quasiparticles are termed non-equilibrium quasiparticles and present a significant challenge in superconducting quantum circuits since they cannot be removed by reducing the temperature [30].

Quasiparticles in superconducting qubits are a significant source of decoherence since changes in the quasiparticle density gives rise to noise and their presence dissipates energy from the system [4]. In particular, quasiparticle tunnelling is a significant source of decoherence in Josephson Junction based qubits [30]. There have been a few methods created to overcome the effect of quasiparticles in superconductors, with varying degrees of success. These include the introduction of normal metal traps, where a non-superconducting metal is used to trap quasiparticles [31], vortex trapping, where quasiparticles are trapped by vortices induced by an external magnetic field[32], and other methods [30][33].

3.1.3 Kinetic Inductance

The inductance of a resonant circuit consists of a geometrical inductance component and a kinetic inductance component, L_g and L_{kin} respectively, which determine the properties of the microwave circuit [3]. In materials where the mean free path is less than the coherence length predicted by BCS theory, the superfluid density n_s is substantially decreased. In this case, for a given current, Cooper pair velocity is greatly increased and the kinetic energy of the Cooper pairs becomes the dominant factor for the self inductance of the material [34]. Consider a superconducting strip of such a material with a length L, width w and thickness t. The kinetic energy of the Cooper pairs with density n_s and velocity v_s is stored as inductive energy,

$$n_s Lwt\frac{1}{2}(2m)v_s^2 = \frac{1}{2}L_k I^2 \tag{3.5}$$

where L_k is the kinetic inductance of the material and I is the current in the material [34]. In the strip, the current density $J_s = I/wt = n_s(2e)v_s$ [24]. Substituting this into Eq. (3.5), and rearranging, the kinetic inductance is

$$L_k = \frac{2m_e L}{(2e)^2 n_s wt}. \tag{3.6}$$

From Eq. (3.6), we can see that the kinetic inductance and the Cooper pair density are inversely proportional. Therefore,

$$\begin{align}
L_k n_s &= const \tag{3.7}\\
\implies L_k n_s &= (L_k + \delta L_k)(n_s + \delta n_s) \tag{3.8}\\
L_k n_s &= L_k n_s + L_k \delta n_s + n_s \delta L_k + \delta n_s \delta L_k \tag{3.9}\\
\delta L_k(n_s + \delta n_s) &= -L_k \delta n_s \tag{3.10}\\
\delta L_k &= -\frac{\delta n_s}{n_s + \delta n_s} L_k \tag{3.11}
\end{align}$$

When a quasiparticle burst occurs which changes the density of Cooper pairs, high kinetic inductance materials exhibit a larger change in the kinetic inductance than lower kinetic inductance materials.

According to BCS theory of superconductors, the total kinetic inductance of a superconducting wire is related to the properties of the material by by

$$L_{\text{kin,tot}} = NL^{\square}_{\text{kin}} = N\frac{1}{1.76\pi}\frac{\hbar}{k_B T_c}\frac{\rho}{t} \tag{3.12}$$

where $L^{\square}_{\text{kin}}$ is the kinetic sheet inductance of the wire, $\rho = R_S t$ is the room temperature resistivity of the wire, R_s is the sheet resistance, N is the ratio of length to width and t is the thickness of the wire. We choose granular aluminium as the material for our resonators because its high kinetic inductance and high quality factor makes it sensitive to quasiparticle bursts. [3].

3.2 Transmission Parameter for resonators

Our resonators are places in a waveguide and are probed using microwave pulses. This set-up can be modelled as a circuit, as seen in figure 3.1.

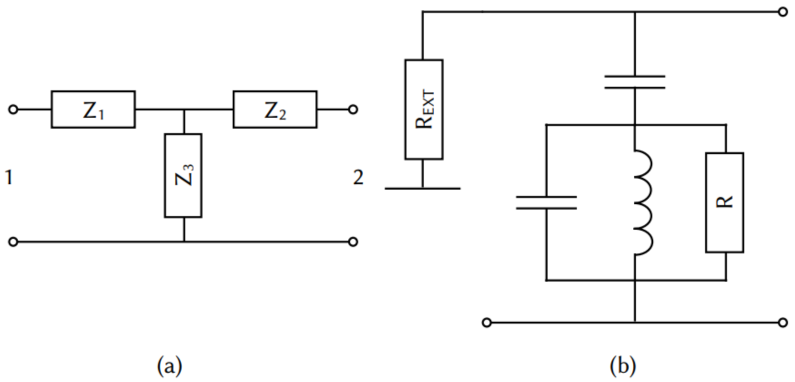

FIGURE 3.1: Model of a resonator coupled to a waveguide in a notch port set-up. (a) Circuit with general impedance's, where Z_1 and Z_2 represent transmission line coupling and Z_3 represents the resonator. (b) The same circuit using specific circuit elements [2]

The resonator itself is modelled as an LCR oscillator, with internal losses modelled by R. The resonator is capacitively coupled to the electric field in the waveguide, here modelled as a transmission line. The coupling to the resonator and the losses due to the waveguide is modelled with a capacitor and a resistor with resistance R_{ext} respectively. By making the assumptions that the resonator has a single pole, we can derive the transmission parameter for the resonator in the waveguide following the procedure outlines in [2]. Complex impedance's can each be modelled generally as

$$Z = \frac{a + bi}{c + d\omega i} \tag{3.13}$$

where a, b, c and d are complex numbers. Since we are interested in the $S21$ parameter, we write it generally as

$$S_{21} = \frac{g + i\omega h}{c + d\omega i}, \tag{3.14}$$

where g and h are also complex numbers. First, we assume that the resonator is lossless. A lossless resonator shows no transmission at resonance, due to a $\pi/2$ phase shift from the resonator [2]. This is encoded into the transmission spectrum by introducing a resonant frequency ω_r,

$$S_{21} = \frac{ih(\omega - \omega_r)}{c + d\omega i}. \tag{3.15}$$

Since the overall magnitude and phase is unimportant, the expression is simplified to

$$S_{21} = \frac{i(\omega - \omega_r)}{k + i\omega}. \tag{3.16}$$

From experiment, the measured resonance is of the form of a Lorentzian [2]. This is accomplished by defining $k = \kappa + i\omega r$, where κ is a complex number. Substituting k into Eq. (3.16), the transmission parameter becomes

$$S_{21} = \frac{i(\omega - \omega_r)}{\kappa + i(\omega - \omega_r)} \tag{3.17}$$

$$= 1 - \frac{\kappa}{\kappa + i(\omega - \omega_r)}, \tag{3.18}$$

where second term has the form of a Lorentzian

$$L(x) = \frac{1}{\frac{1}{2}\gamma + i(x - x_0)}. \tag{3.19}$$

Here x_0 is the so called resonant frequency and γ is the full width as half maximum. It is at this point that we reintroduce dissipation into the system as the imaginary component of the resonant frequency [2]. That is, $\omega_r \to \omega_r + i\epsilon$. We can also split κ into its real and imaginary components as $\kappa = \kappa^{Re} + i\kappa^{Im}$. Substituting into Eq. (3.18), the transmission can then be written as,

$$S_{21} = 1 - \frac{\kappa^{Re} + i\kappa^{Im}}{(\kappa^{Re} + \epsilon) + i(\omega - (\omega_r - \kappa^{Im}))}. \tag{3.20}$$

We can see that κ^{Im} acts as an effective change in the resonance frequency for the resonator, therefore we can define then $\omega_r' = \omega_r - \kappa^{Im} = \omega_r + \delta\omega$.

$$S_{21} = 1 - \frac{\kappa^{Re} - i\delta\omega}{(\kappa^{Re} + \epsilon) + i(\omega - \omega_r')}. \tag{3.21}$$

The total quality factor for a Lorentzian is defined as $Q_l = x_0/\gamma$. The total quality factor for our resonator is therefore calculated to be

$$Q_l = \frac{\omega_r'}{2(\kappa^{Re} + \epsilon)}. \tag{3.22}$$

The total quality factor has two interpretations which are logically equivalent [3].

$$Q_l = 2\pi \frac{\text{Average energy stored in the resonator}}{\text{Energy dissipated per cycle}} \tag{3.23}$$
$$= \omega \frac{\text{Total energy stored in the resonator}}{\text{Total power dissipated}}. \tag{3.24}$$

Total quality factor can be split into two components, the internal quality factor and the coupling quality factor, each of which represents some loss mechanisms in the system [3]. They are related to the total quality factor by

$$\frac{1}{Q_l} = \frac{1}{\Re(Q_c)} + \frac{1}{Q_i} \tag{3.25}$$

where Q_c is the complex coupling quality factor and Q_i is the internal quality factor. By comparing Eq. (3.25) and Eq. (3.22), the coupling quality factor and the internal quality factor for the resonator can be found to be

$$\Re(Q_c) = \frac{2\kappa^{Re}}{\omega_r'} \tag{3.26}$$
$$Q_i = \frac{\epsilon}{\omega_r'}. \tag{3.27}$$

Substituting into Eq. (3.21) ,the transmission parameter becomes,

$$S_{21} = 1 - \frac{\frac{\omega_r'}{2\Re(Q_c)} - i\delta\omega}{\left(\frac{\omega_r'}{2\Re(Q_c)} + \frac{\omega_r'}{Q_i}\right) + i(\omega - \omega_r)}. \tag{3.28}$$

Finally, this can be simplified to become

$$S_{21} = 1 - \frac{Q_l/Q_c}{1 + 2iQ_l\left(\frac{\omega - \omega_r'}{\omega_r'}\right)} \tag{3.29}$$
$$= 1 - \frac{\frac{Q_l}{Q_c}e^{i\phi_0}}{1 + 2iQ_l\left(\frac{\omega - \omega_r'}{\omega_r'}\right)} \tag{3.30}$$

where ϕ_0 is the impedance mismatch between Z_1 and Z_2 of the transmission line [2].

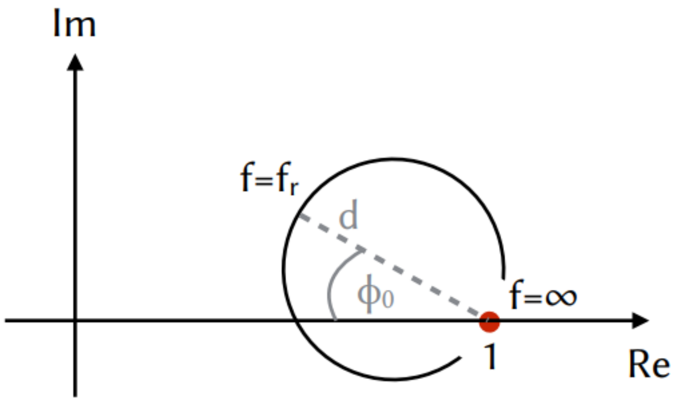

Figure 3.2: Plot of the real and imaginary components of the transmission parameter around the resonant frequency. The circle intersects the real axis at one and is rotated about this point by the impedance mismatch ϕ_0 [2]

3.2.1 Circle Fit Routine

In order to extract parameters from a measured resonance, the circle fitting routing was developed. Here, the convention is changed to use frequency f instead of angular frequency ω. The transmission parameter around the resonance frequency will make a circle in the complex plane, as seen in figure 3.2.

The circle intersects the real axis at $\Re(S_{21}) = 1$, which occurs when the frequency f goes to infinity. As such, this point is called the off-resonance point. The circle is rotated around this point by the impedance mismatch ϕ_0, and the radius of the circle is given by Q_l/Q_c. Since the coupling quality factor Q_c is always greater than the load quality factor Q_l, as seen in Eq. (3.25), the diameter of the circle is smaller than one. So far, we have not considered environmental effects due to the cables. Taking these into consideration, the transmission parameter becomes

$$S_{21}(f) = (ae^{i\alpha}e^{-2\pi i f\tau})S_{21}^{Ideal}(f) \tag{3.31}$$

where S_{21}^{Ideal} is the model derived in equation Eq. (3.30), a and α describe the effects of attenuation and phase shift due to the cables, and τ describes the cable delay. The way that the circle is altered by these effects is shown in figure 3.3. The cable delay causes the phase of the transmission to increase linearly with the frequency. Since the signal takes the same path from source to detector throughout the entire measurement, τ is a constant . The attenuation and phase shift causes the circle to be translated such that the off resonant point sits at (a, α) in polar coordinates, and the diameter of the circle is increased by a factor of a [2]. This can be seen in figure 3.3. The circle fitting routine is divided into a number of subroutines designed to extract certain parameters and allow for others to be more easily extracted.

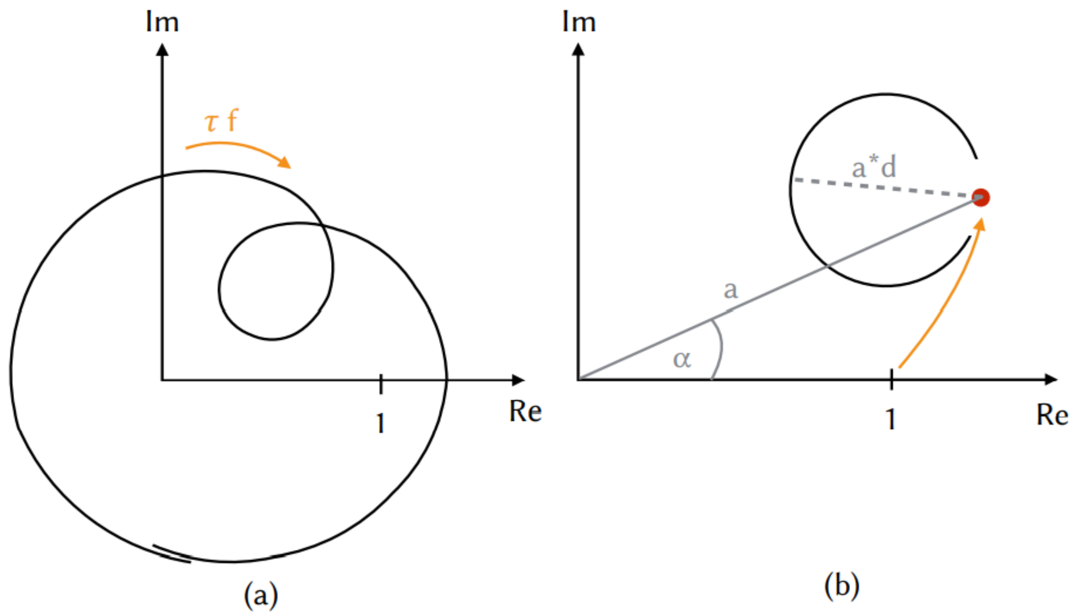

FIGURE 3.3: (a) Effect of cable delay on the circle. Cable delay adds a linear dependence between the phase and the frequency. (b) Effect of the additional attenuation on the circle, not taking into account cable delay [2]

Removing the Cable delay

The effects of the cable delay will cause a linear slope in the phase with respect to the frequency, shown in figure 3.4. A linear regression can be applied to the phase and the cable delay can be determined. The effect of the cable delay is then removed from the signal.

Finding the off-resonant point

The off-resonant point can be found by transforming to a coordinate frame with the origin in the center of the circle. At this point, the phase at each point in the circle can be described by

$$\theta(f) = -\theta_0 + 2\arctan\left(2Q_l\left(1 - \frac{f}{f_r}\right)\right) \tag{3.32}$$

where f_r is the resonance frequency and θ_0 is the phase on resonance. Eq. (3.32) is fitted to the circle, and each of the parameters are extracted. Since the off-resonant point is directly opposite to the phase at resonance, $\theta_\infty = \theta_0 + \pi$. Now that the off-resonance point is found, the attenuation a and the phase shift α can be found by transforming the circle back to where the off-resonance point sits on $1 + 0i$. Since the fit is highly sensitive to noise, the parameters Q_l and f_r are not used in the final results [3]. Instead, they are determined in the final steps of the routine.

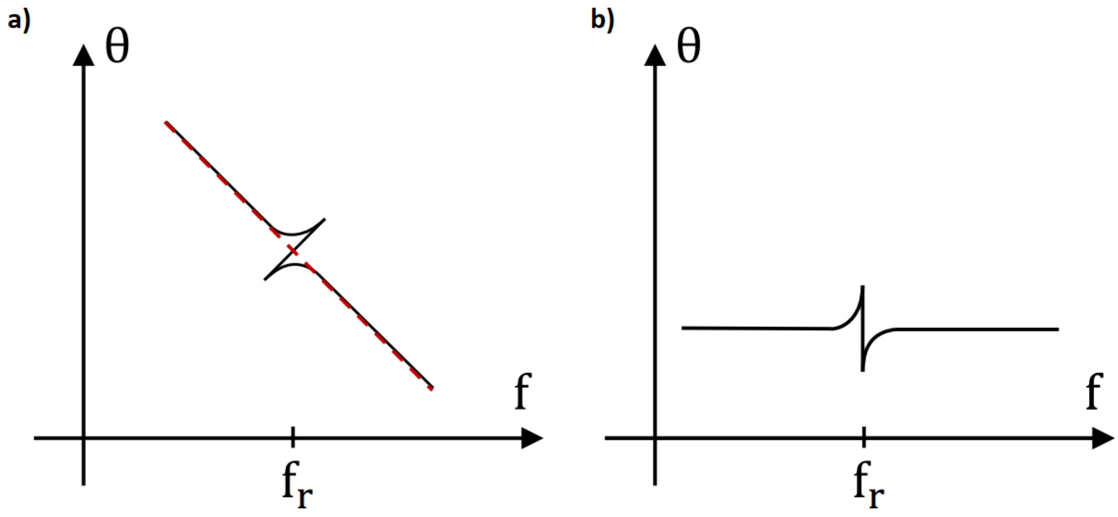

Figure 3.4: Effects of the cable delay on the phase of the transmission parameter. a) Measured phase includes cable delay which is linearly related to the frequency. b) Phase once the cable delay had been removed. Adapted from [3]

Circle Normalisation

Since the circle was fitted for in order to perform the previous step, all that is needed to be done here is to resize the circle by dividing the diameter by a and finding the impedance mismatch ϕ_0. The impedance mismatch is given by

$$\sin(\phi_0) = \frac{y_c}{d} \tag{3.33}$$

where d is the corrected diameter of the circle, and y_c is the imaginary coordinate of the center of the circle.

Final Steps

Now, the resonance frequency f_r and the load quality factor Q_l are extracted by fitting to the magnitude of the S_{21} parameter, which provides more robust results than the fit using Eq. (3.32). The coupling and internal quality factors are now determined. The real part of Q_c is found as

$$\Re(Q_c) = \frac{Q_l}{d\cos\phi_0} \tag{3.34}$$

and the internal quality factor is determined using Eq. (3.25) [3].

4
Literature Review

4.1 Introduction

In this chapter, I will summarise three main papers which have driven our work over the last year. Our work was inspired by (Cardani et al., 2020), which used granular aluminium resonators to provide strong evidence that non-equilibrium quasiparticles are generated by high energy cosmic radiation impacting the chip on which superconducting devices sit on. (Vepsäläinen et al., 2020) supports this conclusion with simulations and experiment. (McEwan et al. 2021) demonstrates the extent of the damage that correlated quasiparticle bursts have on qubit coherence times and for devices that need to work together on the same chip. These three papers show that reducing the quasiparticle burst rate in superconducting devices is extremely important to improving qubit coherence times, as well as allowing for quantum error correction protocols.

4.2 Reducing the impact of radioactivity on quantum circuits in a deep-underground facility (Cardani et al., 2020)

This paper released in May 2020 showed that non-equilibrium quasiparticle bursts were correlated across multiple devices on the same chip and that the quasiparticle burst rate could be reduced by adding extra shielding around the cryostats that the devices were placed in [4]. The paper used three 20 nm thick granular aluminium resonators on a sapphire chip to detect quasiparticle bursts over time. The chip was placed in a copper waveguide and cooled to a temperature on the order of 50 mK. The experiments were performed in three different set-ups. The first set-up took place in Karlsruhe, Germany, in 2018, where they secured the chip to the waveguide using silver paste and used copper and aluminium shields. The second was done in Rome, Italy, where two measurements were done, comparing the effects of using silver paste as opposed to vacuum grease to secure the chip. They showed that using vacuum grease, being less radioactive, radioactive, resulted in a lower quasiparticle burst rate across all three resonators. The third set-up took place in the Gran Sasso National Laboratory, Italy. The experiment was set-up 1.4 km below ground, equivalent to 3.6 km of water. At first, the experiment is run with the cryostat being surrounded by lead bricks. The quasiparticle burst rate was approximately 50 times smaller than in the other two set-ups. The second measurement removed the lead bricks and saw an increase in the quasiparticle burst rate. The third measurement added a radioactive ThO_2 source and recorded a quasiparticle burst rate higher than the other two set-ups. The quasiparticle burst rate and the internal quality factor at the single-photon limit are shown for each resonator and for each experiment in figure 4.1a.

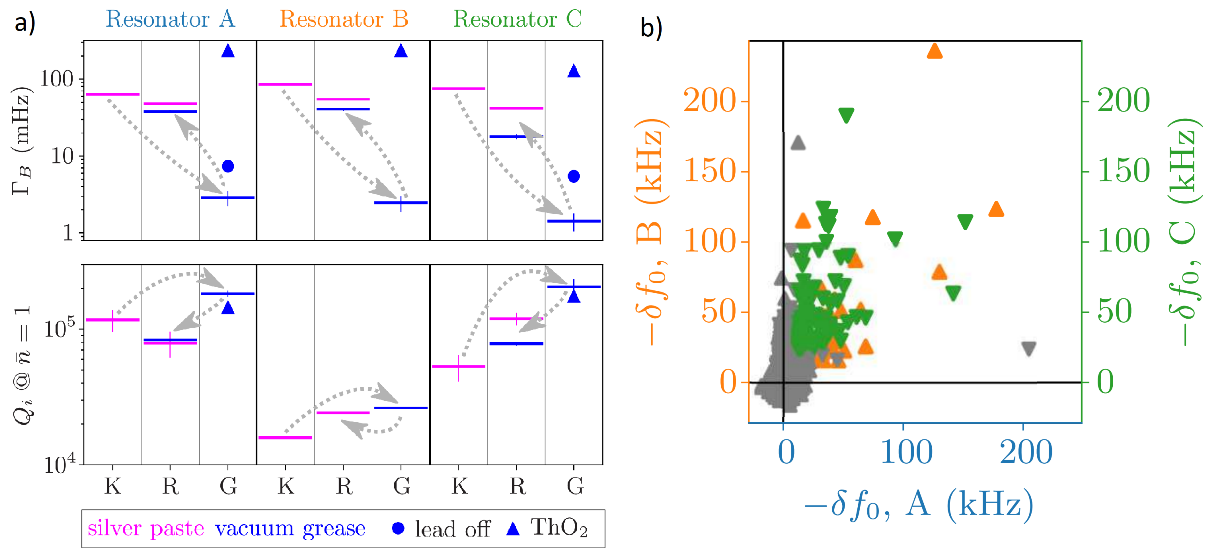

FIGURE 4.1: a) Quasiparticle burst rate and internal quality factor for each of the three resonators used in the experiments. K, R and G correspond to the locations of the experiment (Karlsruhe, Rome and Gran Sasso respectively). b) Quasiparticle bursts are directly proportional to the measured frequency shift. The frequency shifts that occur simultaneously in resonator A and resonator B are plotted together using the orange marks. The frequency shifts that occur simultaneously in resonator A and resonator C are plotted together using the green marks. The grey marks correspond to data that is below the statistical threshold to be conclusively labelled a quasiparticle event [4].

As a preliminary result, the paper managed to show that quasiparticles tended to occur in multiple resonators at the same time. Since there must be some process that carries energy to each of the resonators in order for this to occur, they use this as evidence that quasiparticle bursts are being created due o phonons in the device substrate. The correlation plot for quasiparticle bursts across the resonators is shown in figure 4.1b.

From these two results, the paper concludes that non-equilibrium quasiparticles are created when high energy radiation, likely cosmic radiation, makes it through the cryostat shielding and interacts with the device substrate. This creates phonons that propagate through the chip and deliver enough energy to devices to cause Cooper pairs to break and generate quasiparticles [4].

4.3 Impact of ionising radiation on superconducting qubit coherence (Vepsäläinen et al., 2020)

This paper, released in August of 2020, introduced a quantitative model for quasiparticle dynamics and aimed to provide experimental validation of the effect of ionising radiation on superconducting qubits. The paper first proposes a model for the quasiparticle dynamics based on the recombination rate and generation rate. The paper then discusses two main experiments. The first experiment placed a piece of radioactive copper, ^{64}Cu, which had been irradiated with neutrons in the MIT Nuclear Reactor Laboratory, resulting in a radioactive source with a half-life of 12.5 hrs. The copper was mounted in a cavity 3.3 mm above two transmon qubits. The Γ_1 relaxation rate was measured for each qubit over a period of 400 hours and is shown in figure 4.2.

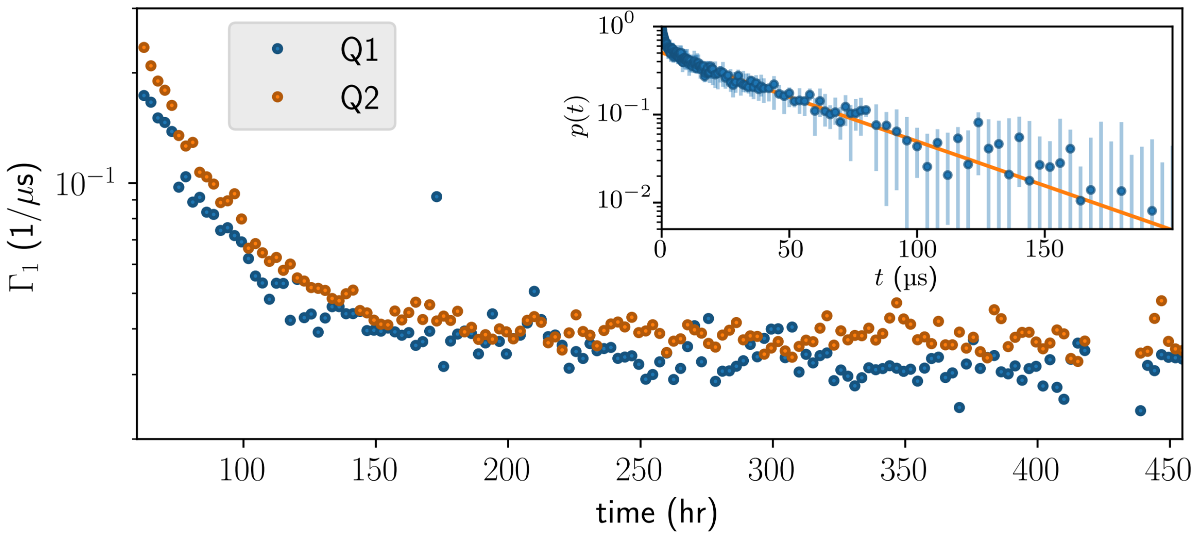

FIGURE 4.2: The relaxation rates measured over time of two transmon qubits in a cavity with a radiation source with a half life of 12.5 hours. The inset shown an example T_1 measurement which works by finding the probability of the qubit being in the excited state after being left for a time t [5]

.

The Γ_1 rate decreased from $1/5.7\ \mu s^{-1}$ to $1/35\ \mu s^{-1}$ in qubit 1, with a similar decrease in qubit 2, due to the copper becoming less radioactive over time. In addition to radiation affecting the qubit relaxation time, the resonance frequencies of the readout resonators shifted due to quasiparticle induced changes to their kinetic inductance, which was consistent with their proposed model. After calibrating their model using the radioactive copper, they removed the radioactive copper from the system. The power density that results in quasiparticles is then categorised into two terms. The internal power density accounted for power sources internal to the dilution refrigerator, such as infrared photons from higher temperature stages. The external power density is due to ionising radiation from outside the dilution refrigerator. The second experiment aimed to determine the effects of external power sources on the quasiparticle density in the qubits. They mounted 10 cm lead bricks on a scissor lift so that the shield could be easily raised and lowered around the cryostat. They then alternately lifted and lowered the shield and measured the Γ_1 relaxation rates for the qubits. They managed to show that the shield had statistically improved the T_1 relaxation times of their qubits. The main results from this paper are that they directly relate the quasiparticle density in superconducting devices to the intensity of the radiation interacting with the chip [5].

4.4 Resolving catastrophic error bursts from cosmic rays in large arrays of superconducting qubits (McEwen et al., 2021)

This paper manages to show that correlated quasiparticle events for devices on the same chip can be seen to originate from one localised event [6]. The paper uses 26 of the 53 available flux tunable transmon qubits on a Google Sycamore Processor to measure quasiparticle bursts. The qubits are uncoupled to each of their neighbours and are spaced 1mm apart. They introduce a method that they call rapid repetitive correlated sampling to measure quasiparticle bursts. Essentially, they excite all of the qubits to the excited state and leave them idle for a short time of 1 μs before simultaneously measuring the state of each qubit. Any qubit that has decayed to the ground state is recorded as an error. This cycle is repeated at regular intervals of 100 μs for extended periods of time. Quasiparticle bursts can be seen on the chip by an increase in the number of simultaneous qubit errors on the chip. This technique for detecting quasiparticles was so accurate that in figure 4.3 (a), they showed the leading edge of a quasiparticle burst as it is formed. Figure 4.3 (b) shows the quasiparticle burst on a larger time scale, including the start of the recombination.

They show that high energy events produce bursts of errors that affect an entire patch on the processor and can last up to the order of a few seconds. Using fine time-resolved measurements, they also show that quasiparticles events are initially localised but spread over the chip. A heat map of the qubit failure rate averaged over 300 μs windows for four time slices is shown in figure 4.4, and shows how quasiparticles bursts are spread over time. This result provides strong evidence that quasiparticles are generated due to high energy radiation impacting the chip [6].

4.5 Discussion

The first paper by (Cardani et al., 2020) showed strong evidence for quasiparticle bursts originating from high energy radiation impacts on the chip on which superconducting devices sit on. Importantly, they showed that they could reduce the quasiparticle burst rate in

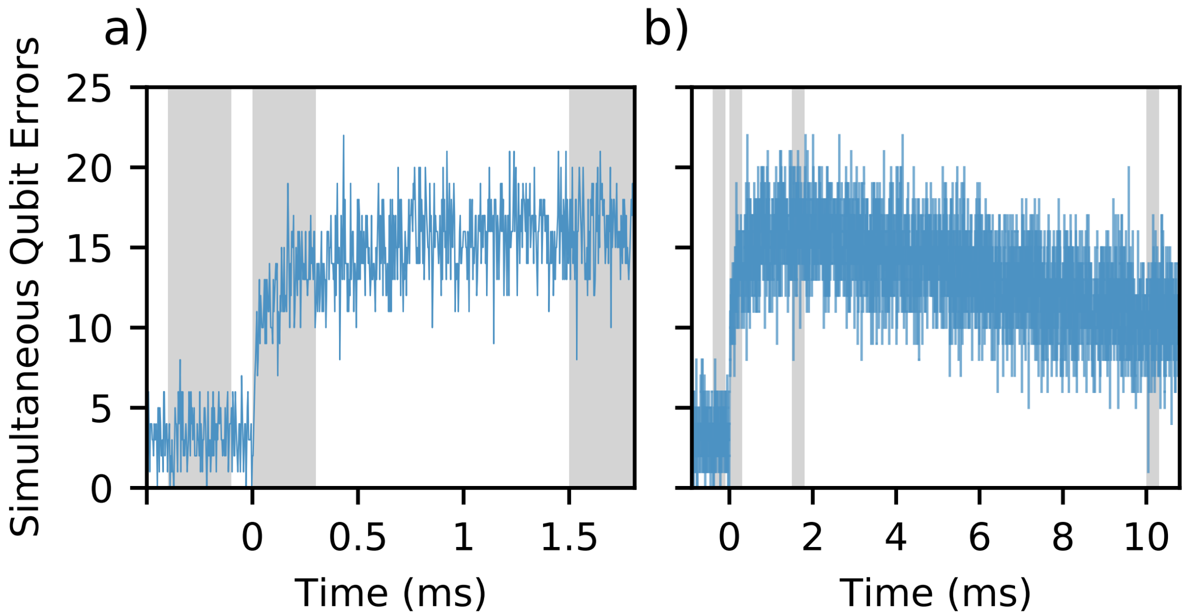

Figure 4.3: Simultaneous qubit failures are used as an accurate detector of quasiparticle bursts across multiple devices on the same chip. (a) shows the rising edge of a quasiparticle burst and (b) shows the rising edge as well as the recombination tail [6].

granular aluminium resonators by shielding the cryostat in a deep underground facility. The second paper (Vepsäläinen et al. 2020) confirmed this theory by first proposing a model for quasiparticle dynamics and then showing that the effects of quasiparticle bursts fit their model. They also showed an increase in the T_1 relaxation time for transmon qubits when they were shielded from the environment. The third paper (McEwen et al., 2021) showed directly that quasiparticle bursts are initially localised and spread across the chip, confirming that high energy radiation impacts the chip and creates phonons that radiate outwards from the impact site and create quasiparticles in the qubits. The idea that phonons in the qubit substrate were the cause of quasiparticle bursts has been a hypothesis for many years. These three papers provide very strong evidence for this hypothesis and have paved the way for future work in reducing the effect of high energy radiation on qubit coherence time, as well as finding ways to reduce correlated qubit failures on the same chip.

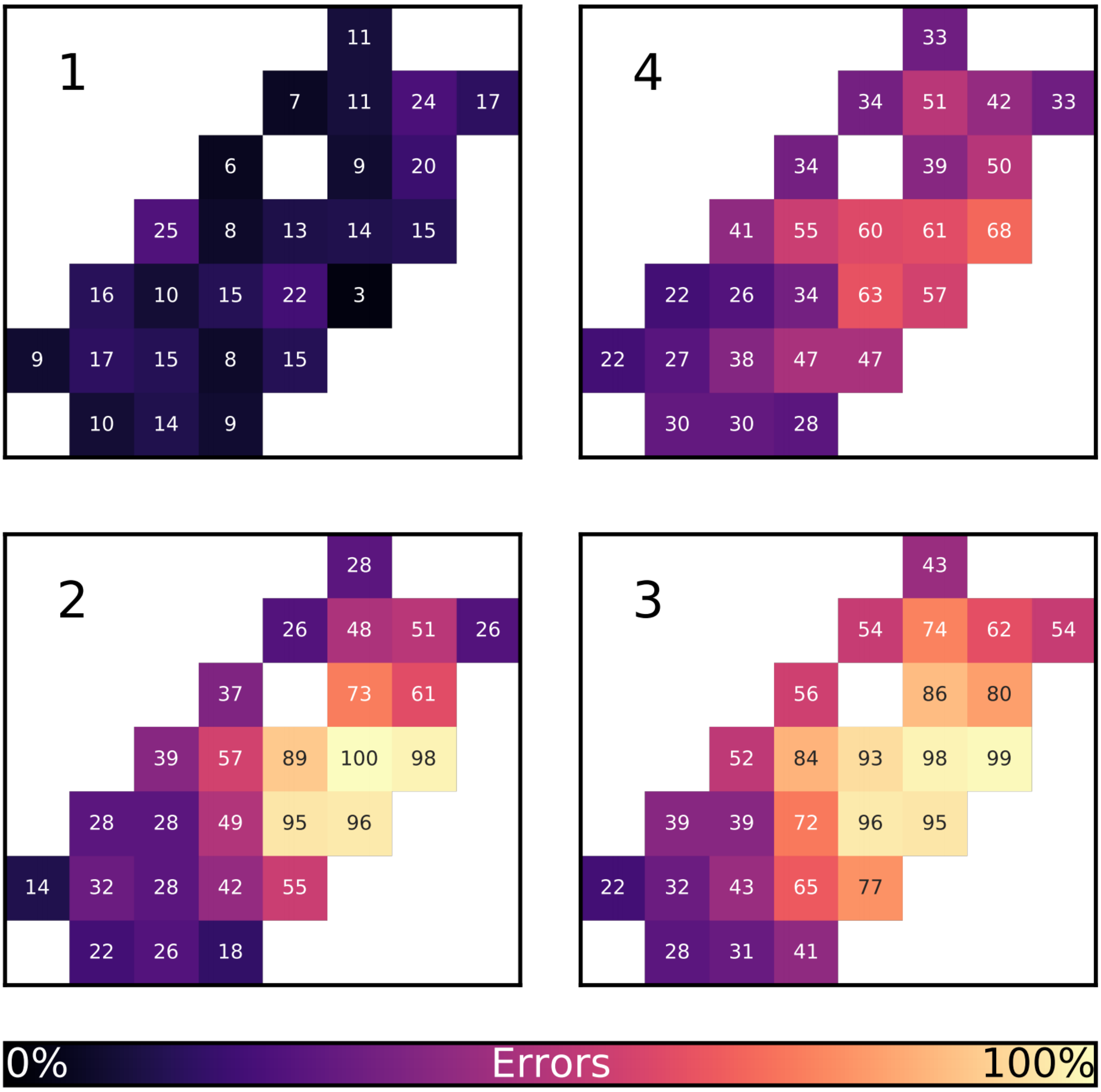

FIGURE 4.4: Heat map of qubit failure rate averaged over 300 μs windows. Time flows from panel 1 to 2 to 3 and ends at panel 4 [6].

5

Finite Element Simulations and Waveguide Design

Our design, similar to the design used in (Cardani et al., 2020), consists of granular aluminium resonators which are placed in a waveguide. The resonators are fabricated onto a silicon chip and are cooled down to superconducting temperature. Before committing to fabrication and ordering components, we needed to understand how the waveguide would interact with the resonators. For this, we use Ansys Electronics HFSS, which is a commercial finite element solver for electromagnetic fields in 3D structures. We based the resonator on one of the resonators used in (Cardani et al., 2020), as ours will be similar. Later resonator designs were simulated and confirmed to be suitable, but the simulation and fabrication of those designs are beyond the scope of this thesis.

5.1 Finite Element Simulations

The finite element solver used in Ansys HFSS assigns a tetrahedral mesh over a 3D structure drawn by the user. At each point in the mesh, Maxwell's equations are solved iteratively until consecutive solutions converge. In this way, the application is able to find an accurate distribution of the electric and magnetic fields over the system. For our designs, we use two simulation routines, the eigenmode solution and the driven mode solution. The eigenmode solution solves for all eigenmodes and cavity resonances in the system, allowing us to easily identify the resonators resonance frequency. We can then do a driven mode solution, in which the user assigns one or more wave-ports to act as sources and detectors of the electromagnetic fields. In this way, we can simulate a measurement of the resonator and ensure that the results are as we expect them to be.

5.1.1 Set-up

We have chosen aluminium SMA couplers to pass our signal through the waveguide. The couplers have internal dimensions of 25.9 ± 0.1 $mm \times 13.0 \pm 0.1$ mm, which corresponds to a cutoff frequency of 5.79 GHz. The dimensions of the simulated waveguide are chosen accordingly, with a length of 80 mm chosen as it is long enough that the signal wouldn't be significantly affected by the size of the waveguide and small enough to fit into the cryostat. The waveguide is drawn in Ansys as a 3d box. The longest edges of the waveguide are modelled as aluminium by assigning a finite conductivity boundary condition. Aluminium was chosen as it would best thermalise with the couplers. The chip is drawn into Ansys as a 3D box with dimensions of 13 $mm \times 8$ $mm \times 0.33$ mm, where the width and the thickness match the chip used in (Cardani et al., 2020) and the height is chosen to match the waveguide designs. The chip is then given the conductance properties of silicon. The resonator itself is drawn as a 2d structure on the center of the chip, with a length and width of 600 μm and 10 μm respectively. Since the resonator will be superconducting, it is given zero resistance, and a reactance $\chi = 2\pi f L_{kin}$, where f is the frequency of the field in the waveguide and L_{kin} is the kinetic inductance of granular aluminium, which is approximated to be 2 nH [35]. The full setup is shown in figure 5.1. In order to ensure accurate results,

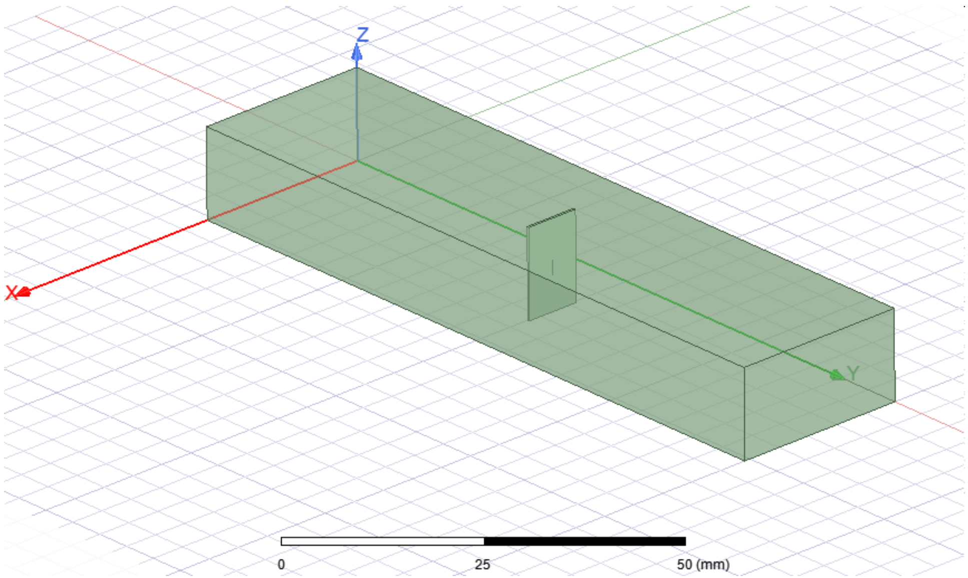

FIGURE 5.1: Ansys design of the experimental setup. Ansys models a superconducting resonator fabricated on a silicon substrate and placed in an aluminium waveguide

the mesh needs to be fine enough to model each component. However, a finer mesh comes at the expense of a significantly increased computation time; hence a trade-off needs to be made. Figure 5.2 shows three examples of a mesh overlay on the resonator. The first mesh doesn't cover enough of the resonator and will likely not show results, while the third overlay will yield results more accurately than what is required but is computationally expensive. I found that a good trade-off is found by assigning the mesh a maximum length that is half of the width of the resonator.

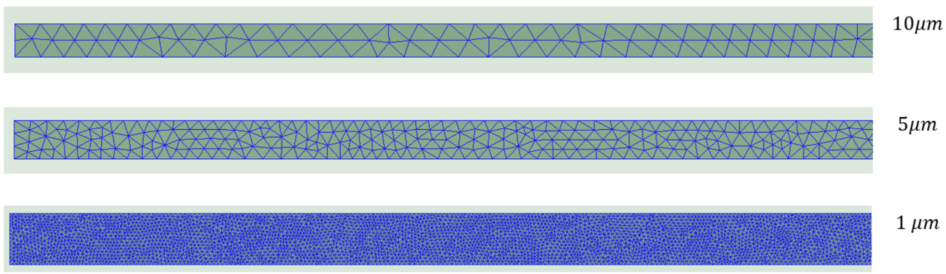

Figure 5.2: Three examples of mesh settings that could be applied to the resonator. The first is not fine enough to properly resolve the resonator, while the third is too computationally expensive. The second resonator shows a mesh setting that worked well for our setup with a reasonable computation time

5.1.2 Eigenmode Solutions

The eigenmode solver calculates all the eigenfrequencies which may occur in the system. The first eigenmode is plotted in figure 5.3. Though the waveguide is modelled such that

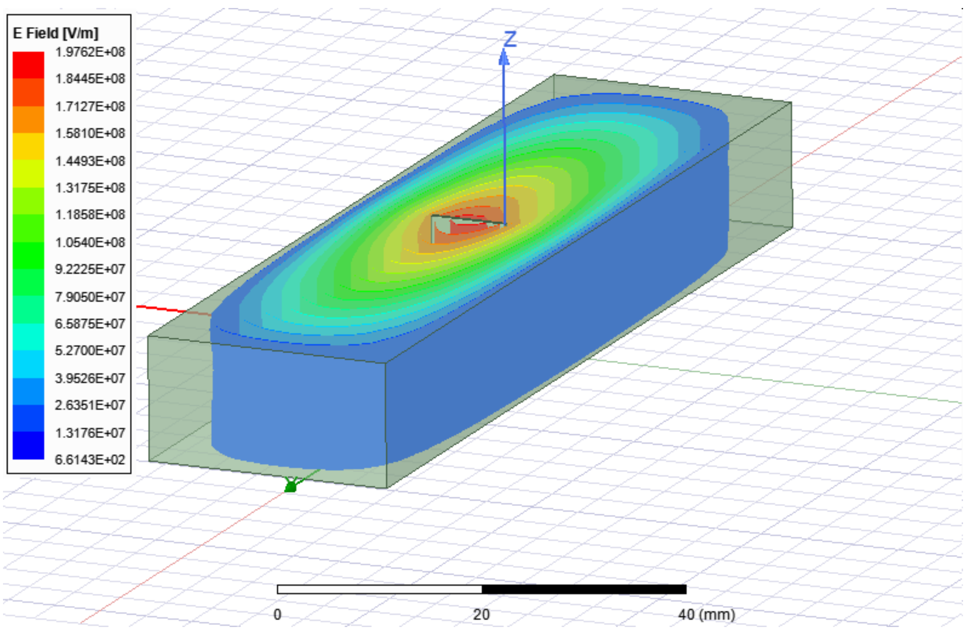

Figure 5.3: Plot of the field distribution for the first eigenmode solution. High energy regions are plotted in red as per the legend

the ends are left unassigned, the eigenmode solver will still treat the waveguide as a cavity. Therefore, most of the eigenmode solutions found will not be physically realisable. The eigenmode solution will, however, calculate the resonance frequency for the resonator and can be identified by plotting the field distribution for each resonance. It is worth mentioning that once the eigenmode corresponding to the resonator is found, the eigenmode solution presents an easy to configure solution for seeing how parameters, such as the length of the

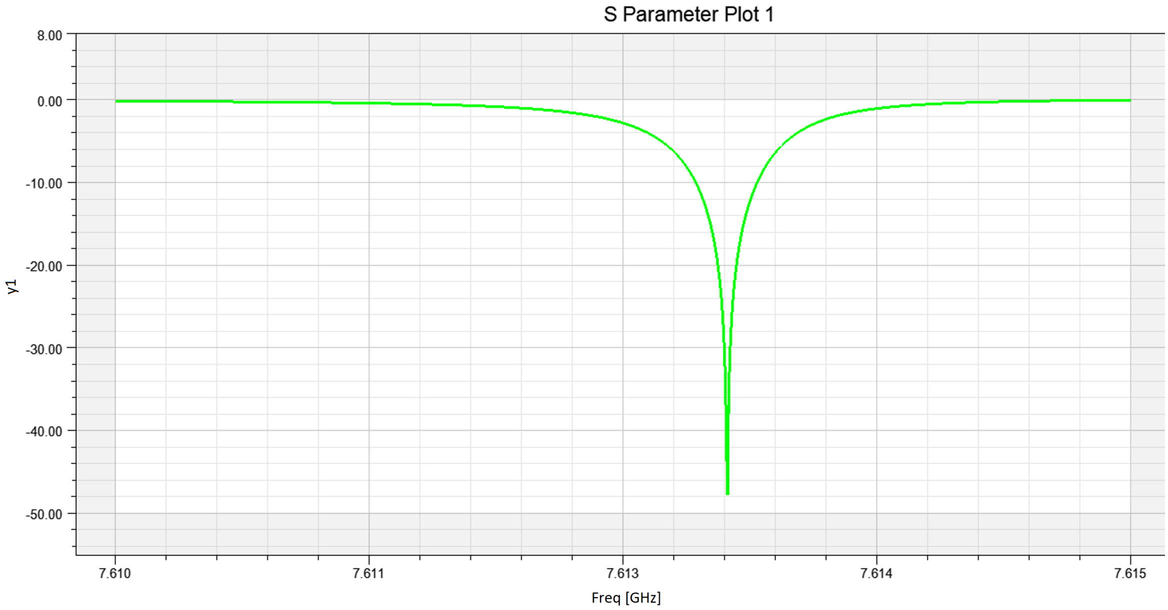

FIGURE 5.4: Plot of the S_{21} parameter generated from the driven mode solution. The resonator has a resonance frequency around 7.61 GHz and a quality factor of 14300

waveguide, affects the resonant frequency. To ensure the accuracy of the simulations, we monitor the convergence metric, which quantifies how different consecutive simulations are from each other. We lower the convergence threshold to $S = 0.005$ and set the minimum number of passes to be at least five, where at least three of these five runs converging below the threshold. From the eigenmode solution, we find that the resonant frequency is approximately 7.2 GHz.

5.1.3 Driven Mode Solutions

Once the resonator eigenfrequency has been identified, we do a driven mode solution. Each end of the waveguide is assigned as a wave port, which will act as both sources and detectors for the microwave signal. For each port, we set the integration line to go from the center to the top edge. While the direction of the integration line is not significant, both ports need the integration lines to be in the same direction. Knowing an approximate resonant frequency from the eigenmode solution, we set the driven mode solution to run from 6.5 GHz to 8.5 GHz, which is a broad scan to try and detect the resonance.

The resonators transmission spectrum is found in the driven mode solution at a frequency of 7.61 GHz which is close to the eigenmode prediction. Since the driven mode solution probes all frequencies around the resonance frequency, the result is more trustworthy. A plot of the raw output is shown in figure

The quality factor for the resonance is estimated from the full width half maximum of the Lorentzian fit to be approximately 14300. This result confirm that our resonators have the correct resonance frequency and quality factor in experiment.

5.2 Waveguide Design

Knowing that the waveguide will result in the resonator having a resonant frequency and quality factor in the range of where we want it to be, we used the dimensions from the simulation. Auto-Cad Inventor is used to design the final waveguide solution. Commercial WR-102 SMA couplers are chosen as they operate within the $7\ GHz - 11\ GHz$ range. Since the chip couples maximally to the electric field in the center of the waveguide, the waveguide was designed in two halves that could be put together. The waveguide is manufactured from aluminium in order to match the material that the couplers are made from. It is important that each side of the waveguide, as well as the SMA couplers on either end, are connected together seamlessly and that the internal chamber is smooth across the connection. Otherwise, imperfections could cause reflections to occur in the waveguide that would disrupt the purity of our signal. The final designs for each side of the waveguide are shown in figure 5.5.

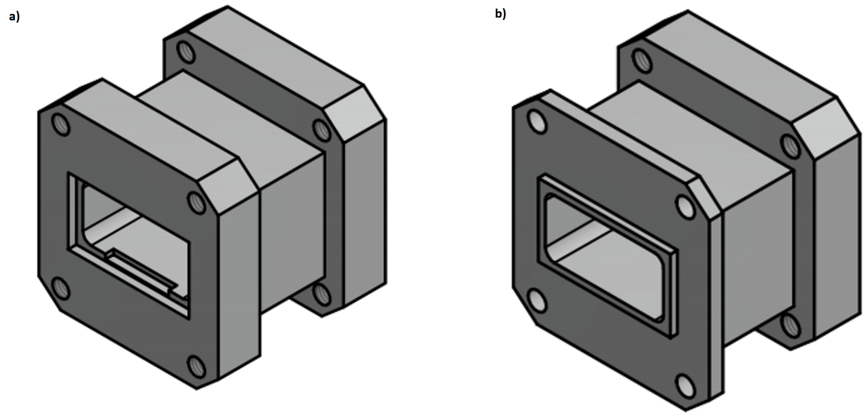

Figure 5.5: CAD designs for each half of the waveguide. a) The left hand image shows the side of the waveguide with a designated slot for the chip. b) The right hand image shows the side of the waveguide with a lip that slots comfortably into the other side

In addition to ensuring the quality of the waveguide, we had to find a way to mount the waveguide to the cryostat in a way that ensures that the system would thermalise well. The setup (the waveguide and the couplers) was too long to be held horizontally in the cryostat, and each side of the setup consisted of the couplers, which were commercially sources, and hence had no inbuilt mounting solution. We designed what turned out to be an elegant solution, which consisted of a copper plate and a threaded dowel rod that could be screwed into both the plate and the waveguide. The plate would sit on top of the waveguide in thermal contact with the coupler. The final set-up can be seen in figure 6.2.

6

Experimental Methods

This chapter outlines the approach taken to set-up the experiment, characterise the resonators and measure quasiparticle bursts. First, the method that we use to detect quasiparticle bursts is discussed. Then, as a description of the experimental set-up is provided. Last, the experimental method is discussed.

6.1 Detecting quasiparticle bursts

As shown in Chapter 3.1.3, the total kinetic inductance of the resonator is directly proportional to the density of charge carriers in the material, in this case, Cooper pairs. When a quasiparticle burst occurs, there is a decrease in the density of Cooper pairs, which results in a change in the total kinetic inductance. If the kinetic inductance is large, the change in the kinetic inductance will be significant. This large change in the kinetic inductance of the material will also change the resonance frequency of the resonators. The relationship between these variables is described by

$$\delta x_{QP} = 2\frac{\delta L}{L} = -4\frac{\delta f_0}{f_0}, \tag{6.1}$$

where x_{QP} is the change in the quasiparticle density, L and f_0 are the kinetic inductance and the resonance frequency before the quasiparticle burst, and δL and δf_0 are the changes in the kinetic inductance and the resonance frequency resulting from the quasiparticle burst [4]. Quasiparticle bursts can therefore be measured by observing the change in the resonance frequency over time. By measuring the complex transmission parameter over time, the change in resonance frequency can be determined. This is shown in figure 6.1.

There are two ways to determine the change in resonance frequency from the transmission spectrum. The first is by observing the change in the magnitude of the transmission at the sample frequency, and the second is by observing the phase. We choose to monitor the

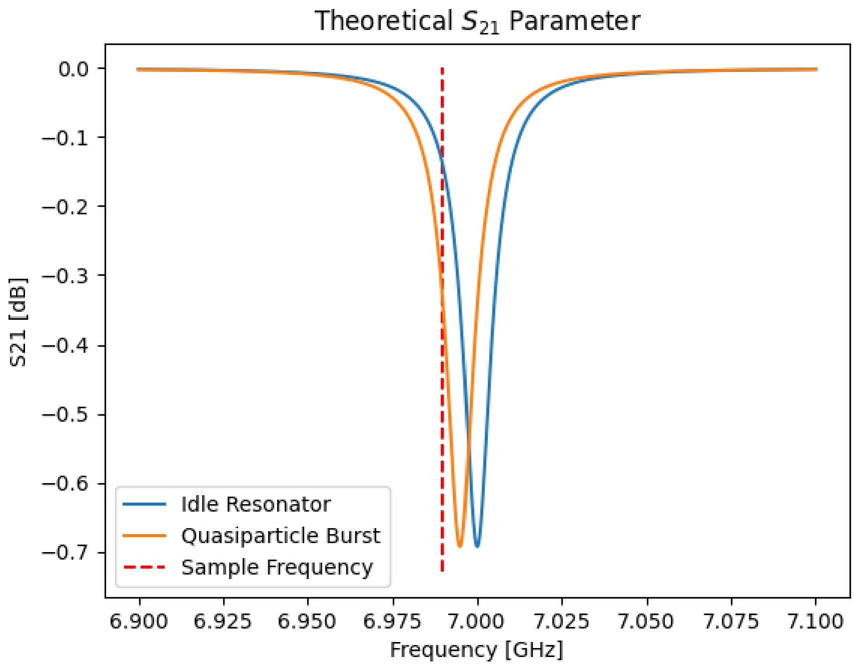

FIGURE 6.1: Diagram demonstrating how the change in resonance frequency due to quasiparticle busts results in a change in the transmission spectrum for the resonators

phase of the transmission due to the convenience of the mathematics, though the choice is arbitrary. By rearranging Eq. (3.32), the resonance frequency can be related to the phase by

$$f_0 = \frac{f}{1 - \frac{1}{2Q_l}\tan\left(\frac{\theta+\theta_0}{2}\right)}. \tag{6.2}$$

The change in the resonance frequency is then related to the change in the phase by

$$f_0 + \delta f_0 = \frac{f}{1 - \frac{1}{2Q_l}\tan\left(\frac{\theta+\delta\theta+\theta_0}{2}\right)} \tag{6.3}$$

$$\implies \delta f_0 = -f_0 + \frac{f}{1 - \frac{1}{2Q_l}\tan\left(\frac{\theta+\delta\theta+\theta_0}{2}\right)}. \tag{6.4}$$

6.2 Experimental Setup

Five granular aluminium resonators were fabricated onto a silicon chip for this experiment. The chip was secured in the middle of the waveguide using vacuum grease, such that the resonators were parallel to the electric field, and therefore maximally coupled. Each side of the waveguide was secured using brass screws and brass nuts. On one end of the waveguide, the SMA coupler is secured using the same brass screws. On the other end of the waveguide, however, the copper mounting component is pressed flat against the SMA coupler, and both components are secured to the waveguide using much longer brass screws and nuts. This was to maximise surface contact between the coupler and the mounting apparatus in order to ensure good thermalisation. The entire system is then secured to the mounting plate of the dilution refrigerator. The dilution refrigerator used is a Bluefors SD dilution refrigerator, which is capable of a final temperature of around 31 mK. A photograph of the final setup can be seen in figure 6.2. Microwave coaxial SMA cabled are used to connect the SMA couplers to the coaxial cables present in the dilution refrigerator. A small cylindrical shield is placed over the waveguide, with the SMA cables carefully bent over the sides. Shields are added to each layer of the cryostat, which act to block both thermal photons from higher temperature layers as well as radiation from the environment surrounding the cryostat. The vacuum container is added to the outer layer of the dilution refrigerator, and the system is depressurised. The system is then left to cool to a final temperature at the lowest stage of 31 mK. The input and output ports of the dilution refrigerator are connected to port 3 and port 4 of a Vector Network Analyser (VNA).

Figure 6.3 shows a schematic of the internal circuitry of the dilution refrigerator and the circuitry around it. The input signal is generated from the VNA and passed through to the input port of the dilution refrigerator. Attenuators in the dilution refrigerator add 70 dBm of attenuation on the input signal, which is important for two reasons. The first reason is to suppress thermal photons from higher temperature stages of the dilution refrigerator from propagating down towards the experiment. The second reason is to lower the power of the raw signal from the VNA, which is too high to be able to resolve the resonators. In addition to the attenuators within the dilution refrigerator, an extra 20 dBm of attenuation is added to the output of the VNA to further weaken the signal. The attenuated signal is passed into the waveguide, where it couples to the resonators and is passed to the output. The output signal is first passed through a low pass filter to remove low energy photons from the signal before they are propagated further. Then, the signal is passed through a superconducting double junction circulator. This device has three ports and will only allow a signal to travel in one direction. By using a 50 Ω termination to block one port, The circulator becomes a two-port device that blocks any signal from being passed into the waveguide through the output port. Thus circulator acts to prevent thermal photons from travelling down the cryostat and into the output port of the waveguide while still allowing a signal from the waveguide to pass through. The waveguide signal is then passed through a high-electron-mobility transistor (HEMT) before being passed to the output of the cryostat. A HEMT is a field-effect transistor that is able to operate effectively at high frequencies [36]. Here, it is used as a cryogenic amplifier before the signal is passed to room temperature amplifiers. Once the signal has been amplified, it is passed to the VNA, where the transmission measurement can be analysed. Due to the simplicity of the setup, not much calibration was needed to detect a signal. Possible impedance mismatches can be easily removed during post-processing.

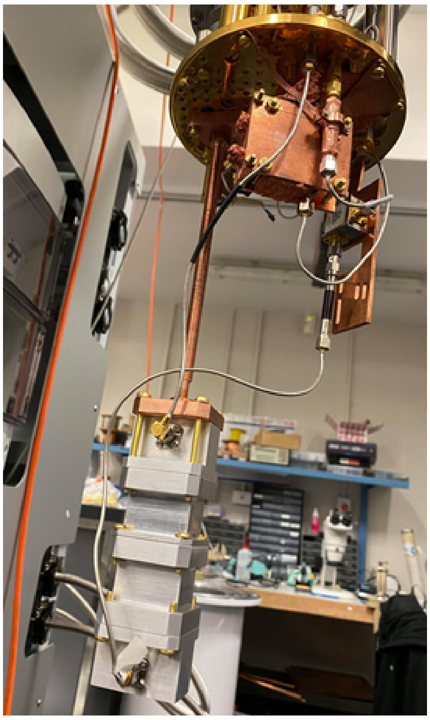

FIGURE 6.2: Final setup for the experiment. The chip is secured inside the waveguide using vacuum grease. The waveguide is secured together with the SMA couplers and mounted onto the mounting plate of the cryostat.

6.3 Data collection

The mixing chamber was allowed to cool to an equilibrium temperature of 31 mK over a period of 2 days. When the experiment was ready, we started off by measuring the transmission spectrum. The scan ranged from 4 GHz to 12 GHz with a bandwidth of 2 kHz. We wanted to make sure that we could see when resonances were occurring, so after accounting for attenuation, the power of this scan was -90 dBm. This was enough power to see a significant signal in the transmission spectrum, but not enough that the losses due to the resonators were drowned out by the input signal. From this profile, we could identify a number of resonances in the transmission spectrum. We now had the task of analysing each one in order to make sure that the resonances were due to the resonators and not due to another loss mechanism present in the system. To do this, we ran scans centered around these resonances at a low power of -120 dBm and a low bandwidth of 500 Hz. The time taken to record one data point is estimated as 1/bandwidth. So, for N data points, the measurement is expected to run for N/bandwidth seconds, making this a time-consuming process. For each resonance, we perform the circle fitting routing discussed in section 3.2.1 using resonator tools, an open-source python library [7]. Specifically, we want to find resonances with high total quality factors. This is because high quality factor resonances will show a greater change in their transmission parameter when a quasiparticle burst occurs, which is demonstrated in section 6.1. A sample circle fit is shown in figure 6.4. Out of the detected resonances, two resonators stood out as being sensitive enough to

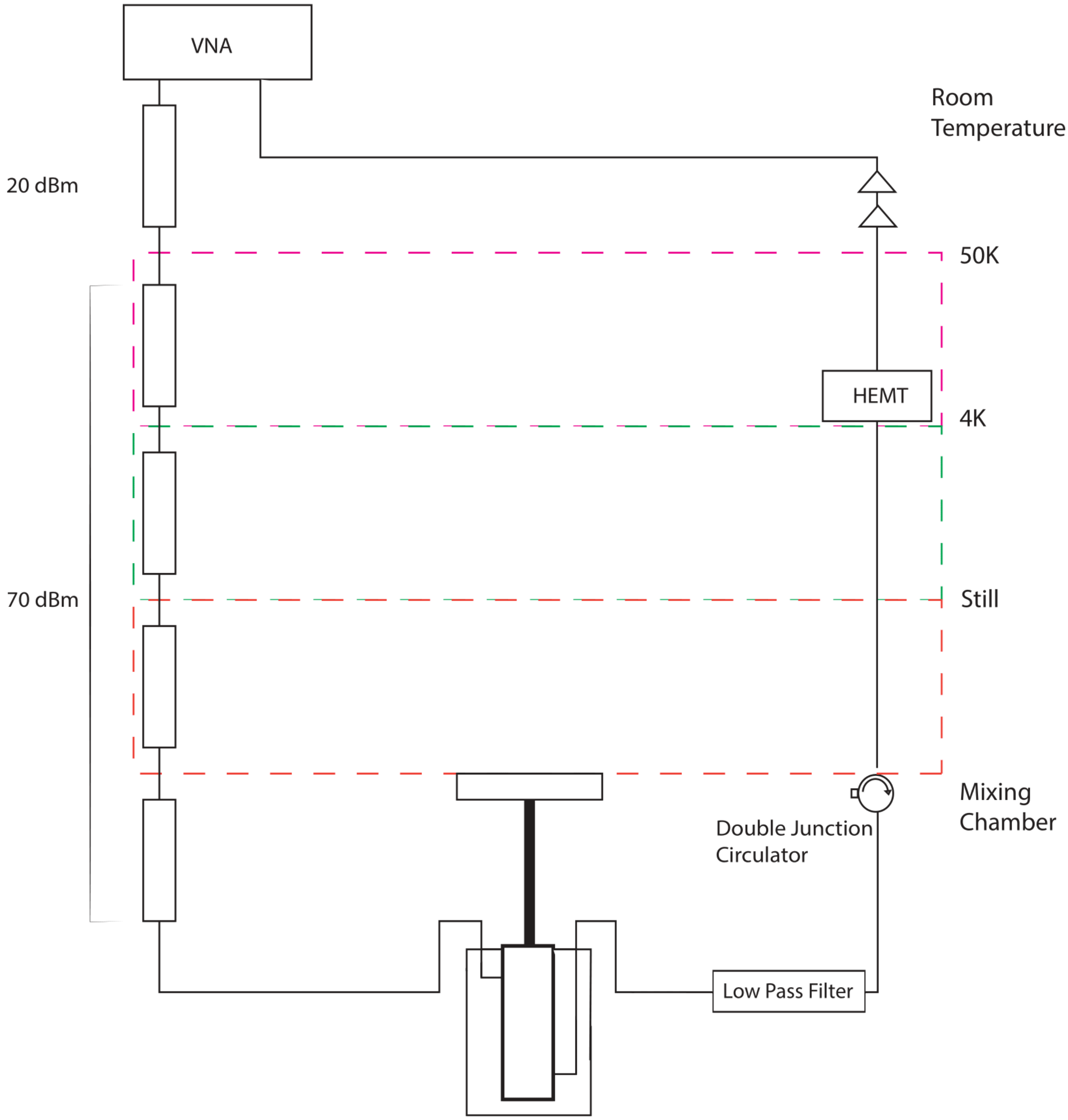

Figure 6.3: Schematic of the internal circuitry of the cryostat. The input signal travels through a total of 90 *dBm* of attenuation before reaching the waveguide. The output signal travels through a double junction circulator and a series of amplifiers before reaching the VNA

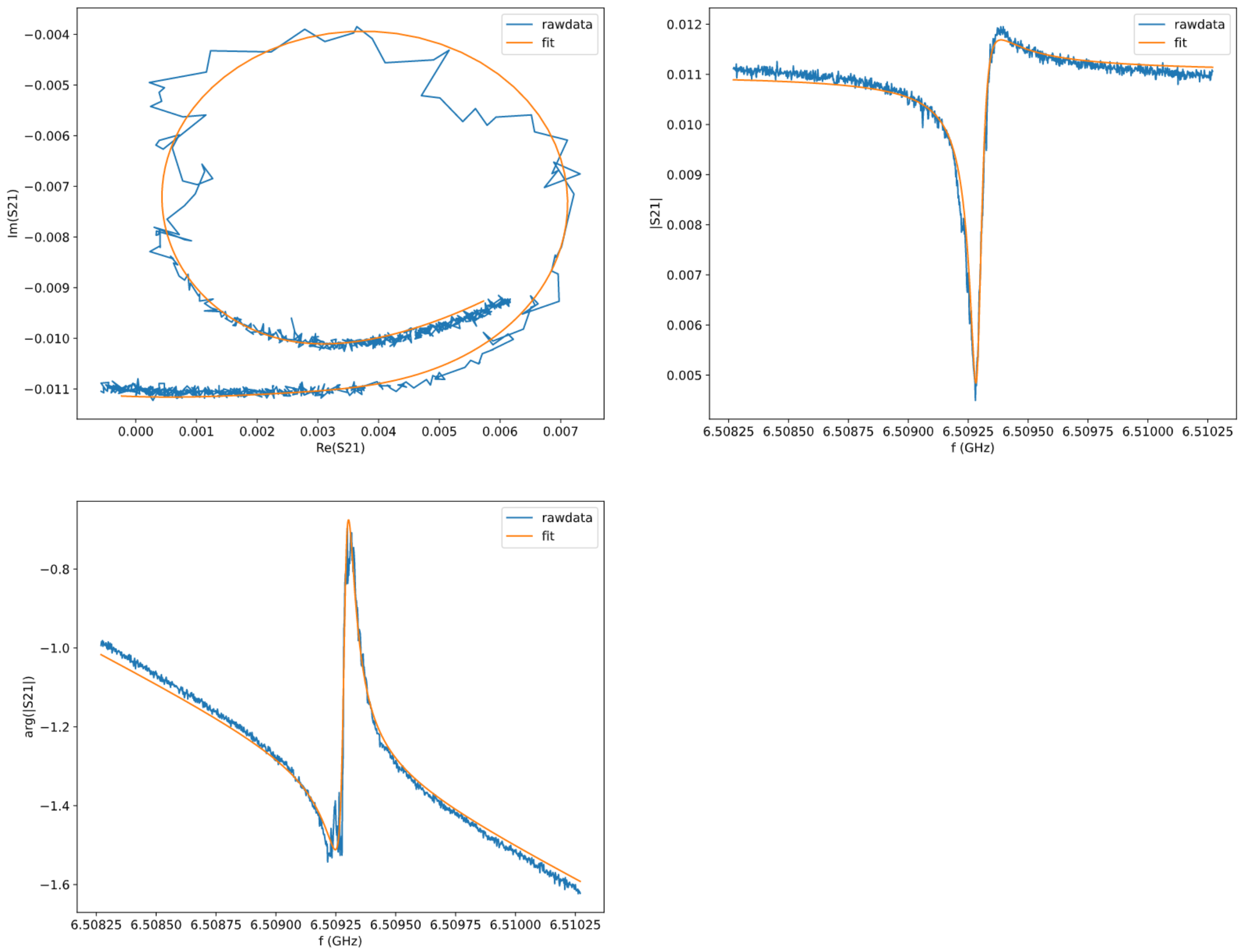

FIGURE 6.4: Example of an output of the circle fit routine developed in the resonator tools library [7]

detect quasiparticles. Due to their high quality factors, they were difficult to spot in the initial scan but were discovered after subsequent investigations. The first resonance had a resonance frequency of 6.509296 GHz and showed a load quality factor of 90906. The second has a resonance frequency of 9.264314 GHz and has a load quality factor of 64000, significantly less than the first resonance. In order to detect quasiparticles, we choose a frequency that is offset from the resonance frequency and monitor the transmission parameter over time. We choose a sampling frequency for the first resonator of 6.509308 GHz and a sampling frequency of 9.264347 GHz. Notice that since we use a bandwidth of 500 Hz for the measurements and fit to this curve to determine the resonant frequency, the frequency measurements are accurate to at least seven significant figures. This is particularly important since our sample frequencies differ from the resonance frequencies by only a few hundred kilohertz. By measuring the transmission parameter at the sampling frequency for the first resonator over time, we are able to detect a quasiparticle event. This is shown in figure 6.5. As you can see, the quasiparticle event shows a sharp change in the transmission parameter followed by a recombination tail, which is the characteristic shape shown in the literature

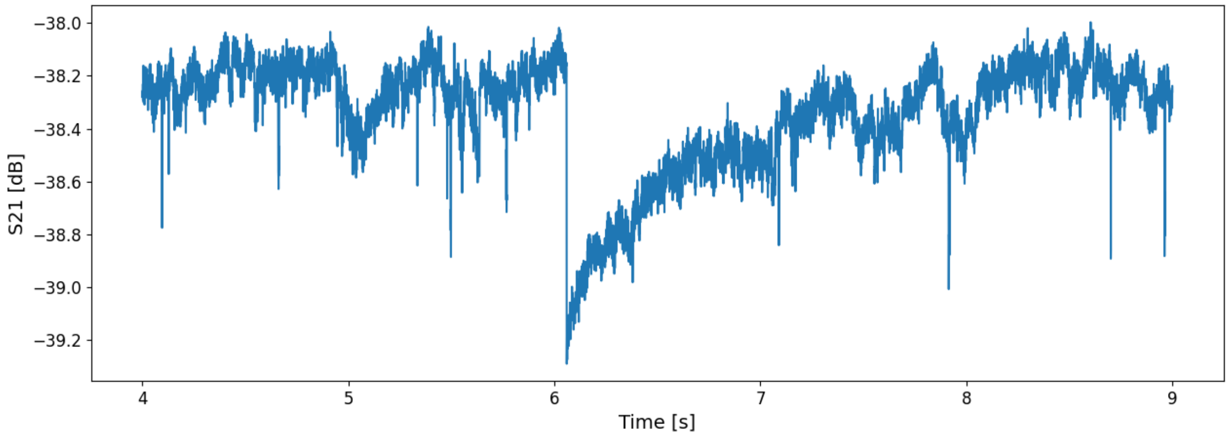

Figure 6.5: Time trace of the transmission parameter during a quasiparticle burst. The burst occurs after six seconds of measurement and lasts for approximately one and a half seconds before all quasiparticles have recombined.

review.

Now, since we aim to measure correlated quasiparticle events, that is, quasiparticle events that occur in multiple devices on the same chip, we alternately measure the transmission parameters for both resonators. The VNA sends out a pulse at the first sample frequency and immediately after at the second sample frequency. The transmission parameter is then recorded for both pulses before the cycle is repeated. The bandwidth for these measurements is set to 1 kHz for fine resolution. Therefore, it takes 2 ms to record a single data point. The delay between consecutive data points depends on how long the signal takes to leave and return to the VNA, which is approximately 20 ms. Using this scheme, the transmission parameters for both resonators are recorded over two runs of 25 hours.

7
Analysis and Results

7.1 Effect of the time of day on resonator phase

Over both 25 hour runs, the change in the transmission was recorded at intervals of approximately 20 ms. If we consider the fact that the quasiparticle burst rate is small in comparison to the sampling rate, fluctuations in the phase are due to noise. The distribution of the phase should then be normally distributed with a mean centered on the phase at the sampling frequency. However, we noticed that the distributions for each resonator in each case consisted of two underlying distributions, a phenomenon most prominent in resonator 1 which has the higher quality factor. After examining the phase over time, a 12 hour window was found in each case where the phase had visibly shifted. Figure 7.1 shows the phase recorded in each resonator for each measurement run, distinguished between points withing the 12 hours and points outside of the 12 hours. I conclude that there is a shift in the phase of the resonators due to solar radiation during the day.

7.2 Determining δf_0

In order to use equation Eq. (6.4) to detect quasiparticle bursts, the phase at resonance (θ_0) needs to be determined. By rearranging equation Eq. (3.32), the phase at resonance is expressed as

$$\theta_0 = -\theta + 2\arctan\left(2Q_l\left(1 - \frac{f}{f_0}\right)\right) \tag{7.1}$$

where θ is the phase at our sample frequency f. Q_l and f_0 are determined accurately by the circle fits for each resonator. Since the distribution of θ is symmetric for each resonator, in each time period of measurement, θ can be estimated from the mean of the phase distribution.

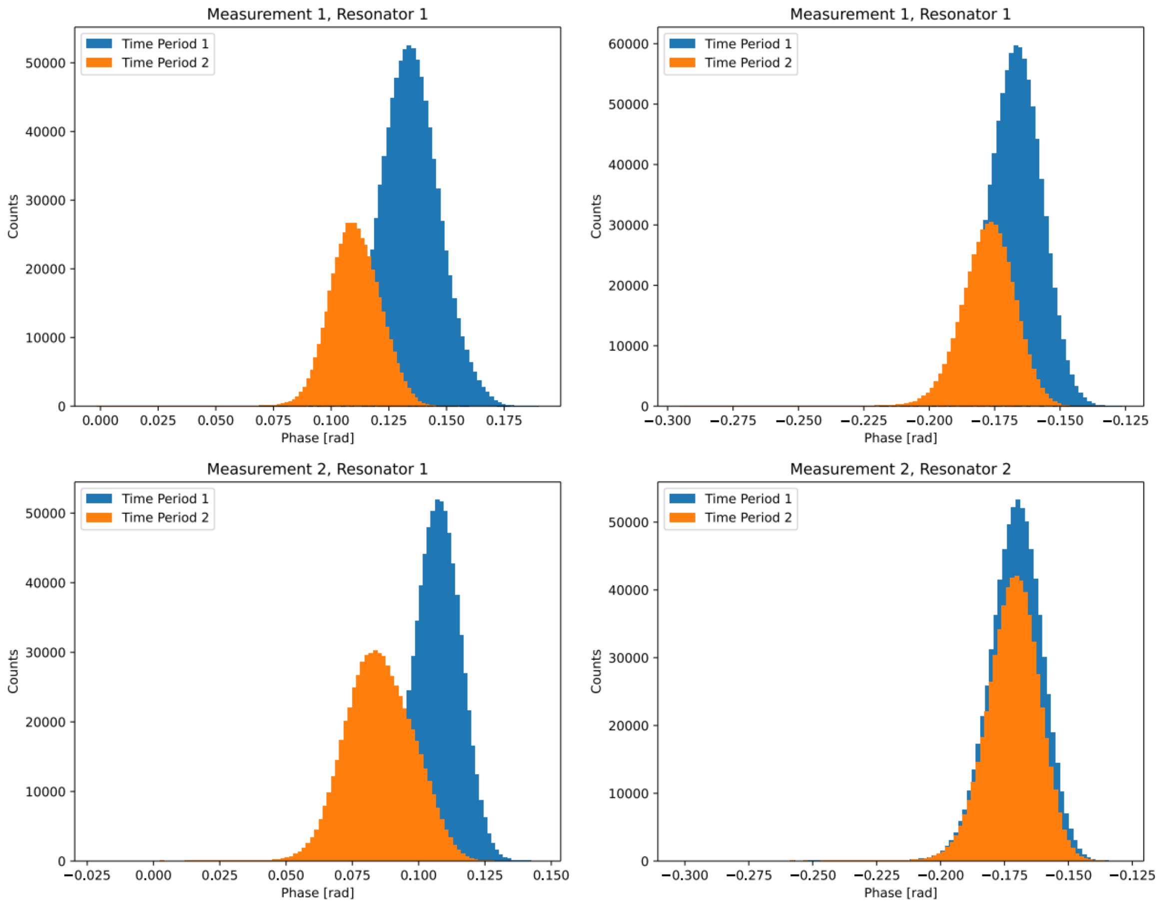

FIGURE 7.1: For each 25 hour measurement run, the phase for both resonators was recorded. In each case, the phase plots are distinguished between periods of 12 hours

Independent measurements of θ_0 are performed in each case. In order to determine the change in resonance frequency, equation Eq. (6.4) is used. The change in phase in each case, $\delta\theta$, is now defined as the change from the mean phase, now denoted $\bar{\theta}$. Denoting the measured phase as θ, $\delta\theta$ is expressed in terms of the data as $\delta\theta = \theta - \bar{\theta}$. Equation Eq. (6.4) is now written as

$$\delta f_0 = -f_0 + \frac{f}{1 - \frac{1}{2Q_l}\tan\left(\frac{\theta+\theta_0}{2}\right)}. \tag{7.2}$$

For each measurement, and for each time period, the change in the resonance frequency is found. The data is then stitched back together so that we have measurements in δf_0 that are not dependent on the time of day.

7.3 Detecting Quasiparticle Bursts

After this analysis, we have four large time series data sets in which to sort through. We need to be able to autonomously and reliably detect quasiparticle bursts in each data set.

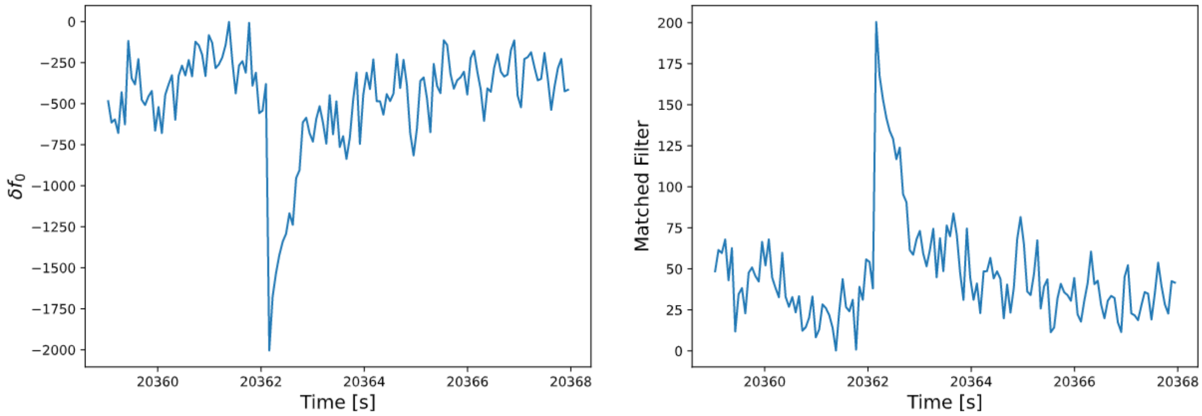

Figure 7.2: Plot of an example quasiparticle burst resulting in a change in the resonance frequency on the left, and the resulting matched filter on the right. The matched filter is designed to detect and amplify a signal in the noise.

To do this, we use a matched filter, an analysis technique designed to optimize the signal to noise ratio [37]. The matched filter is essentially a convolution of the data with a template function. Here, we use the template function

$$T(t) = \begin{cases} a\exp(-(t-t_0)/\tau) & t \geq t_0 \\ 0 & t < t_0 \end{cases} \tag{7.3}$$

which matches the characteristic shape of a quasiparticle burst which starts at time t_0, causes an initial change in frequency of a and has a decay time τ. The filtered signal is therefore given by

$$F(t) = \int_{-\infty}^{\infty} \delta f_0(t')T(t-t')dt'. \tag{7.4}$$

Now, since the template function falls off exponentially, the bounds of the integral can be adjusted to range from t to t_{end} where, t_{end} is when the integrand is effectively negligible. Furthermore, since the data is taken in discrete time steps, the matched filter at time t can be calculated as a Riemann sum, as follows.

$$F(t) \approx \sum_{i=0}^{N_{end}} \delta f_{0,i} T(t-t_i)\Delta t, \tag{7.5}$$

where N_{end} is where the term in the sum becomes negligible, and Δt is the time between consecutive measurements.

Spikes in the filtered data will indicate when the signal matches the template function, and thus will indicate quasiparticle bursts. We detect quasiparticles by recording the peaks in the matched filter which have a signal strength that is at least four standard deviations from the mean signal strength. Including lower signal strength spikes may lead to the inclusion of false positives. Figure 7.2 demonstrates how the matched filter acts on the data around a quasiparticle burst.

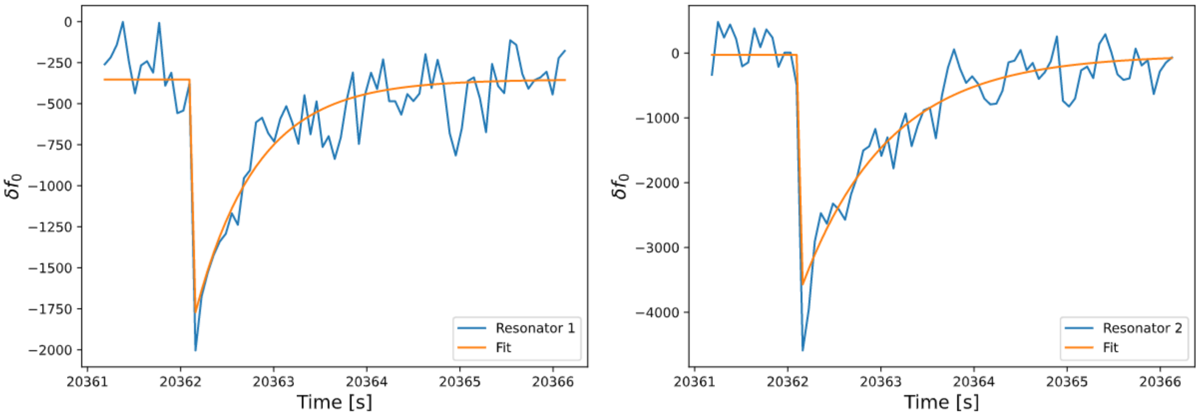

FIGURE 7.3: Plot of the change in resonance frequency due to quasiparticle bursts occurring simultaneously in both resonator 1 and resonator 2. In each case, equation Eq. (7.3) is fit and the decay time was found to be 661 ms in resonator 1 and 930 ms in resonator 2

7.4 Measuring Quasiparticle Burst Rate and Decay Time

The matched filter is run for each resonator over both resonators, and the quasiparticle peaks are found. Since we are looking for correlated quasiparticle bursts occurring due a single event in the substrate, we filter the data to only consider quasiparticle bursts that occur in both resonators at the time time. For each quasiparticle burst detected, the characteristic shape of a quasiparticle burst is fit to the change in the resonance frequency, δf_0 over time. This is the same curve defined in the template function in equation Eq. (7.3). An example correlated quasiparticle bursts and their associated fits are shown in figure 7.3.

From the fits, we can extract the decay times τ for the quasiparticle bursts in each resonator. The results for both measurement runs are combined, since the decay times do not depend on the starting time of the measurement. The quasiparticle decay times for each resonator are shown in figure 7.4. For both resonators, correlated quasiparticle bursts have decay times of up to a second. The mean decay times for quasiparticles in resonator 1 and resonator 2 are found to be 551 ms and 482 ms respectively. Therefore, the quasiparticle bursts in our resonators can last for more than a second in both of our resonators.

The time between quasiparticle bursts is also found, and is plotted on figure 7.5. The distribution of the time between quasiparticle bursts is seen to follow a Poisson distribution, with a Poisson parameter of $\lambda = 615\ s$.

7.5 Discussion

As can be seen from these results, quasiparticle bursts are random and long lasting. We measure that they can occur on average every 615 s, or approximately every 10 min, and have a decay time of up to a second. Since the decay time measures the rate of exponential decay, the quasiparticle density is in fact above the thermal equilibrium expectation for up to a few seconds. The matched filter is a reliable method for increasing the signal to noise ratio of quasiparticle bursts, however it is vulnerable to false positives. These false positives

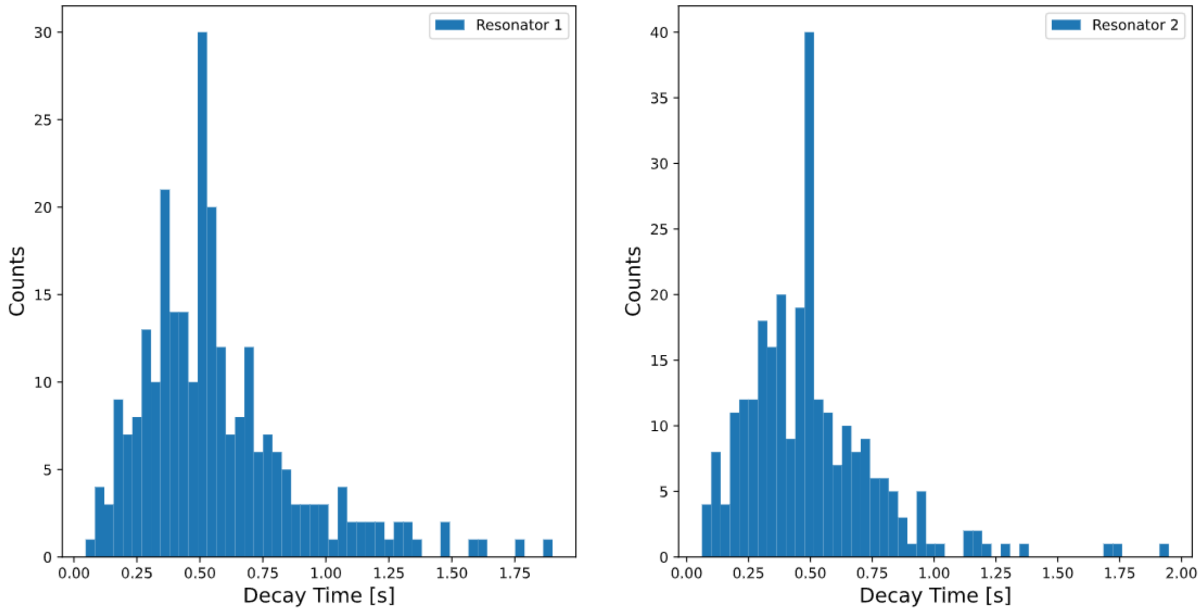

Figure 7.4: Histograms of the quasiparticle decay times in each resonator. Resonator 1 had a mean decay time of 674 ms and resonator 2 had a mean decay time of 510 ms.

are controlled for in two ways. First, since we only consider quasiparticle bursts that occur simultaneously in both resonators, it is less likely that the quasiparticle burst detected in either resonator is a false positive when the other is not. These cases still occur however, and can be detected by the very long decay times assigned by the exponential fit. So, the second way that we control for false positives is by only considering quasiparticle bursts with a decay time of less than 1.5 s. The statistics measured for the mean decay times of the quasiparticles, as well as for the timing between quasiparticle bursts, are approximated by treating the histograms as a accurate representation of the underlying distribution in each case. Quasiparticle bursts are shown to be correlated across multiple devices on the same chip, which agrees with the literature review. Furthermore, the quasiparticle bursts rate is seen to have an underlying distribution given by the continuous Poisson distribution, confirming a result shown in (McEwan et al., 2021)[6].

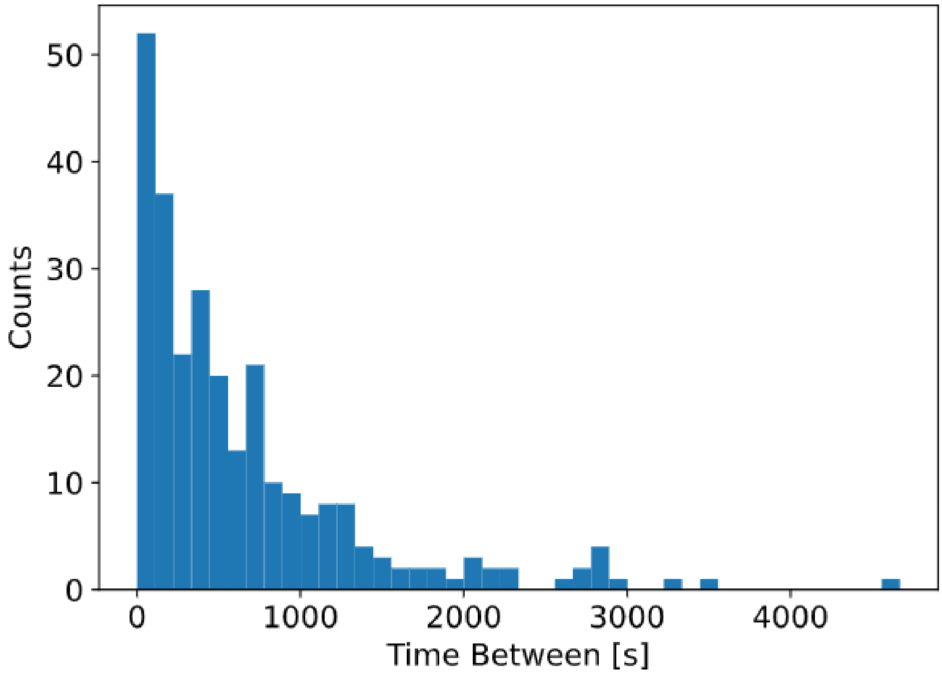

FIGURE 7.5: Histogram of the time between quasiparticle bursts. The time between quasiparticle bursts is a random process that can be described by a Poisson distribution with a Poisson parameter of 615 s

8
Conclusion

High inductance granular aluminium resonators were used to detect correlated quasiparticle bursts in multiple devices on the same chip. First, finite element simulations were done to design an experimental apparatus that could house the chip without detrimental effects to the resonators. The system was placed in a dilution refrigerator and cooled to 31 mK. By alternately probing the transmission parameter for two resonators to determine the resonance frequency, non-equilibrium quasiparticle bursts were detected in both devices simultaneously.

The experimental results of this thesis support recent findings that non-equilibrium quasiparticle bursts are due to high energy radiation impacting the chip on which superconducting devices sit on, causing high energy phonons to propagate through the chip and generate quasiparticles in multiple devices on the chip. Using high inductance granular aluminium resonators to detect quasiparticle bursts, we measured both the quasiparticle burst rate and the quasiparticle recombination rates for correlated quasiparticle events. The quasiparticle burst rate was estimated to be one quasiparticle event every 615 s, and the decay rates where measured to be 674 ms in the first resonator and 510 ms in the second. These quasiparticle events are random and long lasting, and are therefore difficult to control for without designing quasiparticle traps to protect superconducting devices.

This thesis was part of a larger project to prevent quasiparticle bursts from occurring in the first place. Future work that will be done in the Superconducting Quantum Devices Laboratory will be to have granular aluminium resonators fabricated onto a trampoline mechanical resonator structure that will act as a low pass filter for phonons in the substrate. The analysis and experimental design discussed in this thesis will be used in the future for subsequent measurements that will characterise the effectiveness of the trampoline mechanical resonator structure on preventing correlated quasiparticle bursts from occurring.

Removing correlated quasiparticle bursts from superconducting devices could lead to an increase in the coherence times of superconducting qubits, as well as prevent simultaneous qubit errors from occurring in multiple qubits on the same chip.

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